

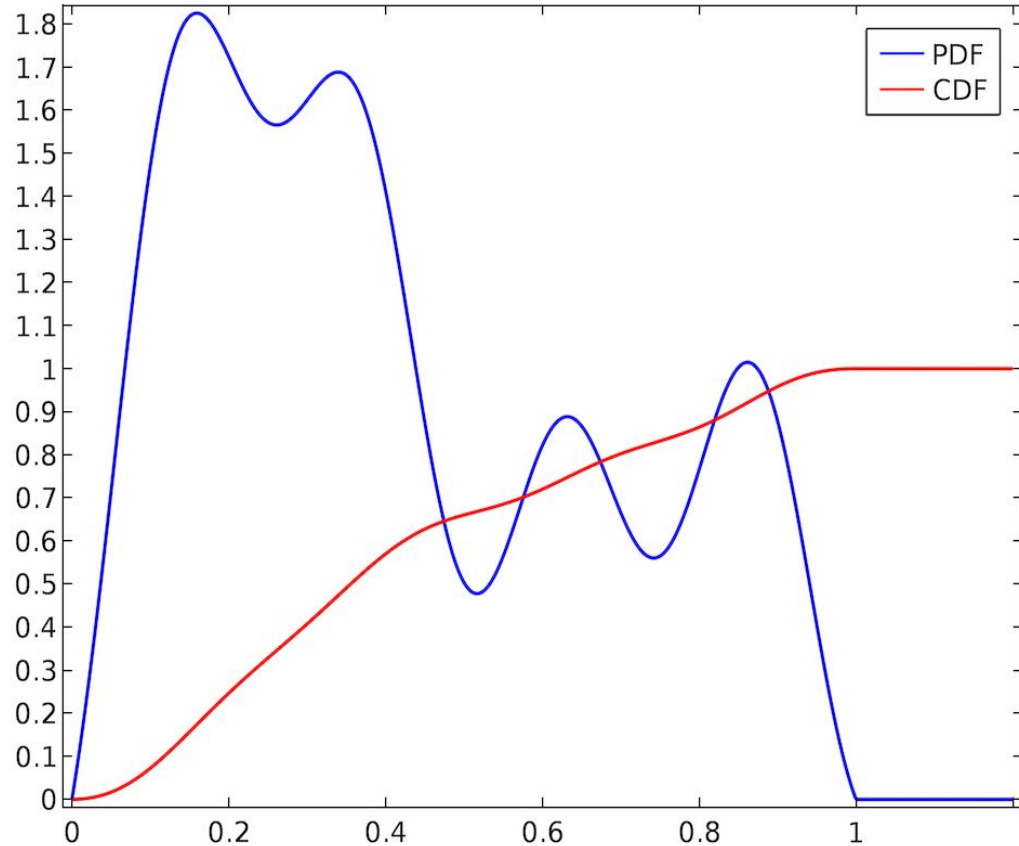
Numerical Methods

Sampling techniques

Kristina Kislyakova, 17.12.2025

Basic Motivation

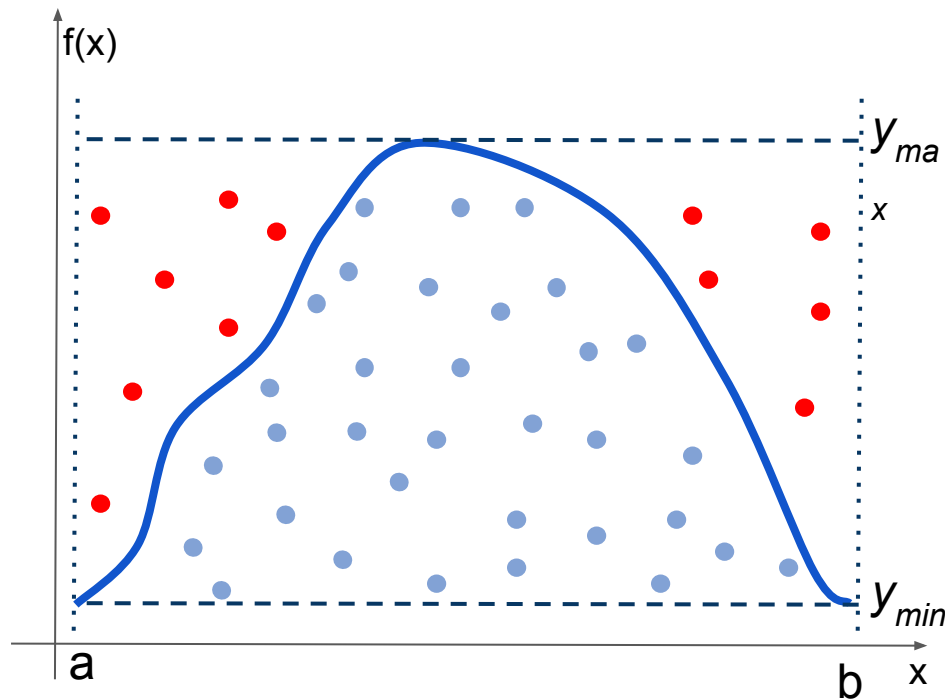
- Often we want random numbers drawn from a probability distribution function (PDF).
- We have seen how to do this for uniform random numbers (*a flat PDF*) or numbers drawn from a Gaussian distribution.
- But how do we get random samples which come from an **arbitrary probability distribution**?



Applications for random sampling

Integration techniques

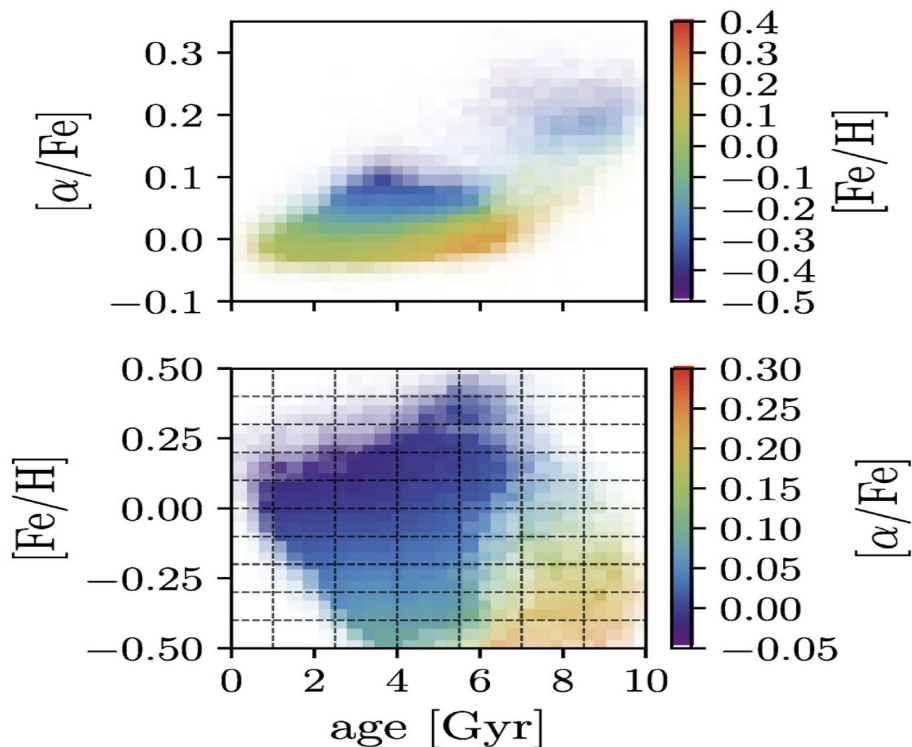
- We saw previously how Monte Carlo methods can be very useful for computing integrals
- This is especially the case for multi-dimensional integrals



Applications for random sampling

Sampling arbitrary distributions

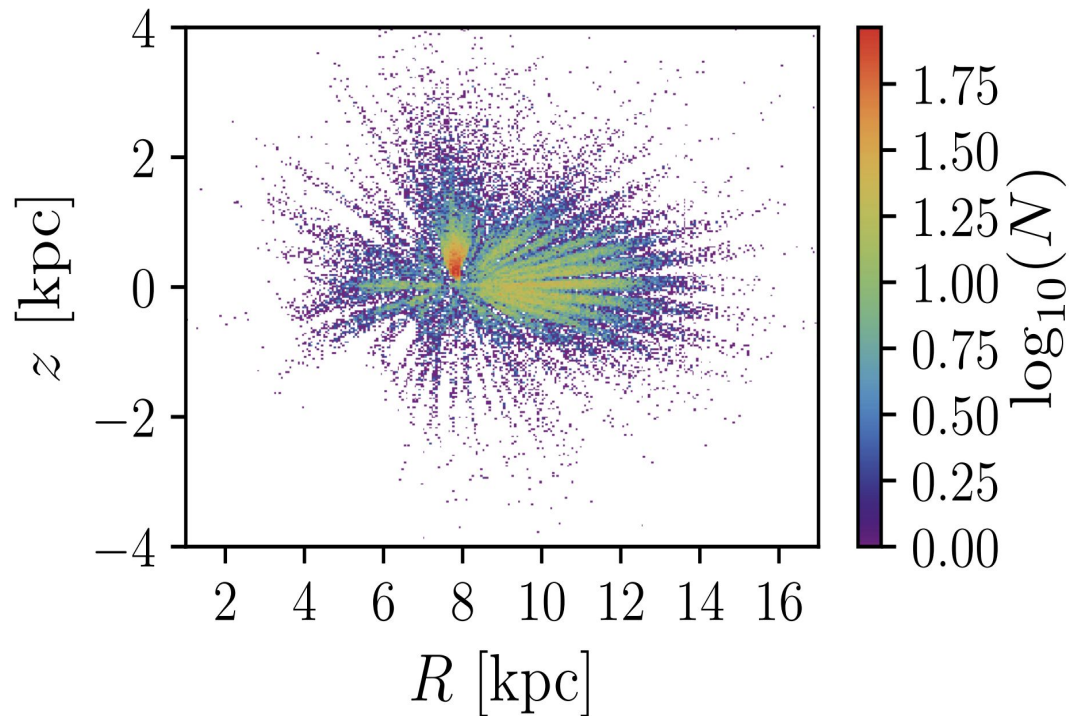
- Many research questions will inevitably require you to produce samples from non-standard distributions
- Such arbitrary distributions will require flexible numerical sampling techniques



Applications for random sampling

Sampling arbitrary distributions

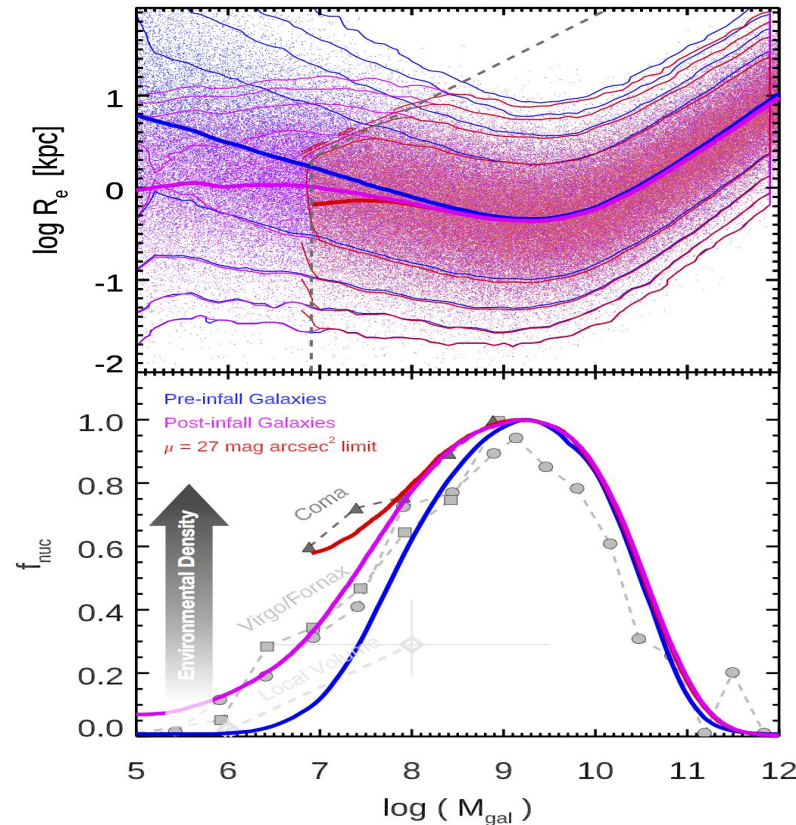
- For example, if you want to infer properties of a larger system from a biased set of observations.
- Understanding a selection function from your survey can often be helped by producing **mock data**.



Applications for random sampling

Sampling arbitrary distributions

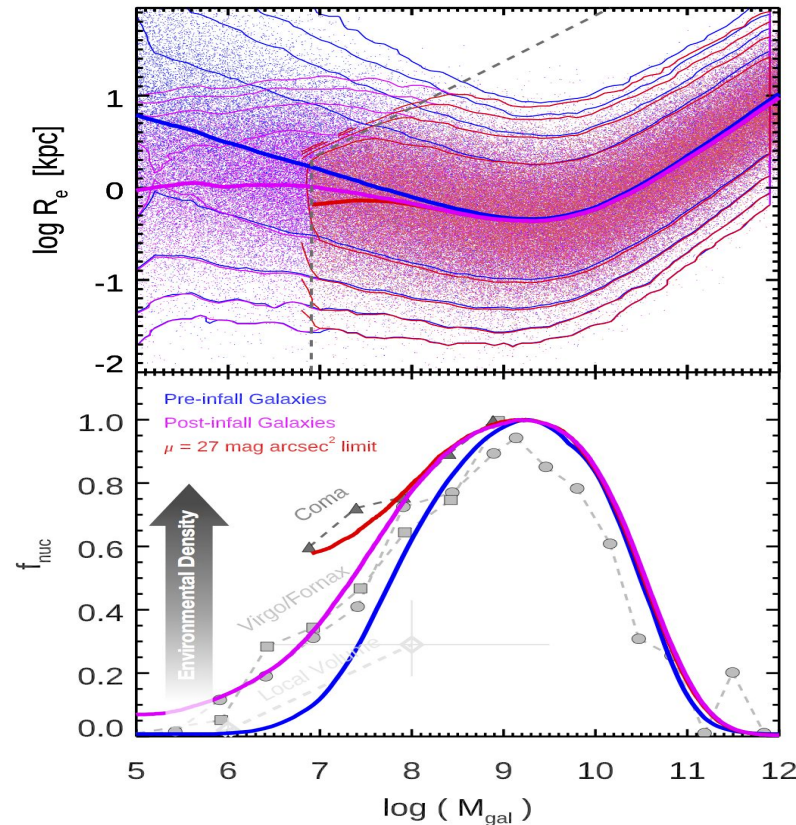
- Basic idea is to start with underlying physical relation representing some system



Applications for random sampling

Sampling arbitrary distributions

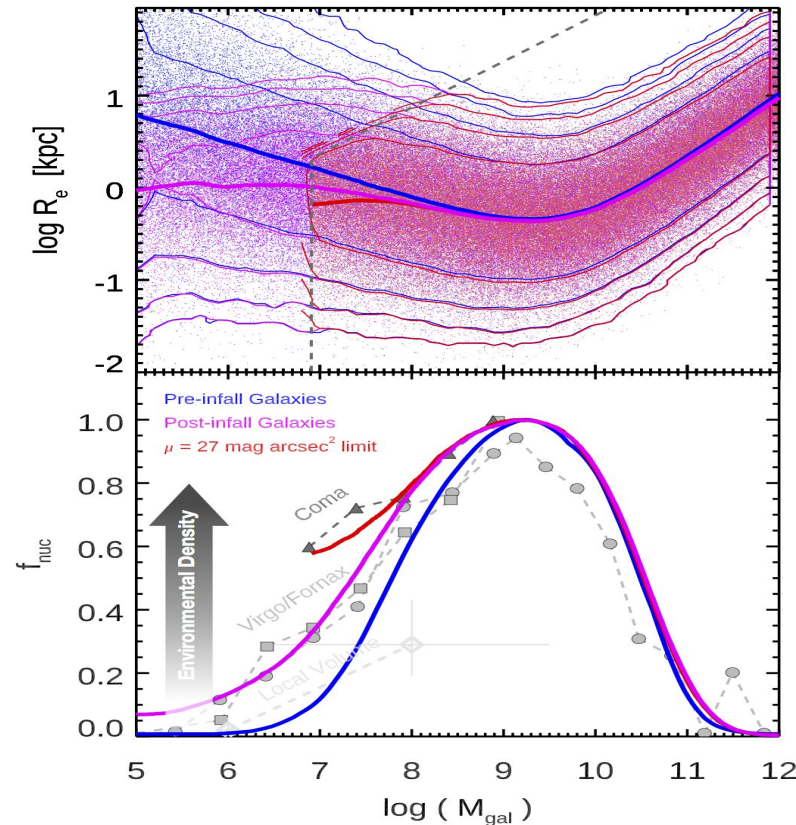
- Basic idea is to start with underlying physical relation representing some system
- Stochastically add in sources of scatter (**intrinsic vs. observational?**)



Applications for random sampling

Sampling arbitrary distributions

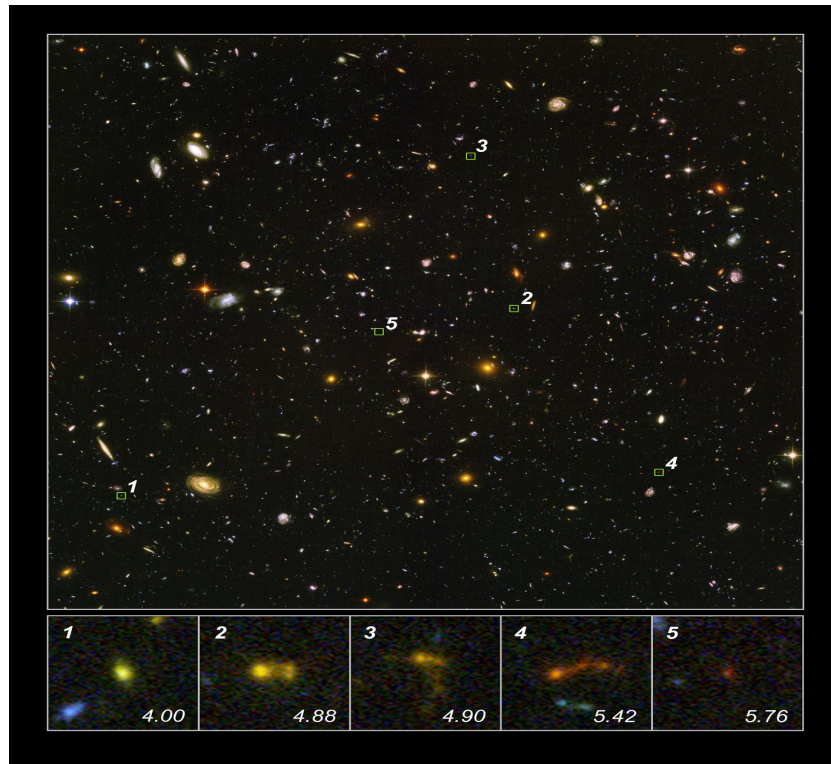
- Basic idea is to start with underlying physical relation representing some system
- Stochastically add in sources of scatter (**intrinsic vs. observational?**)
- Model some systematic effects due to your experiment, or biases within your population



Applications for random sampling

Sensitivity characterization

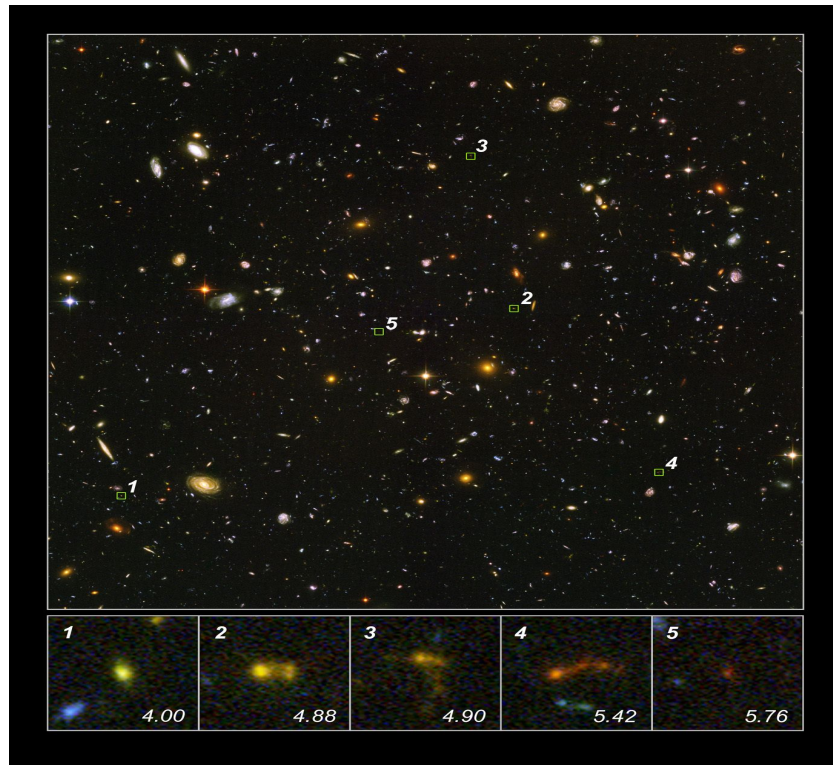
- Another key example is assessing the **sensitivity** of your method/observations



Applications for random sampling

Sensitivity characterization

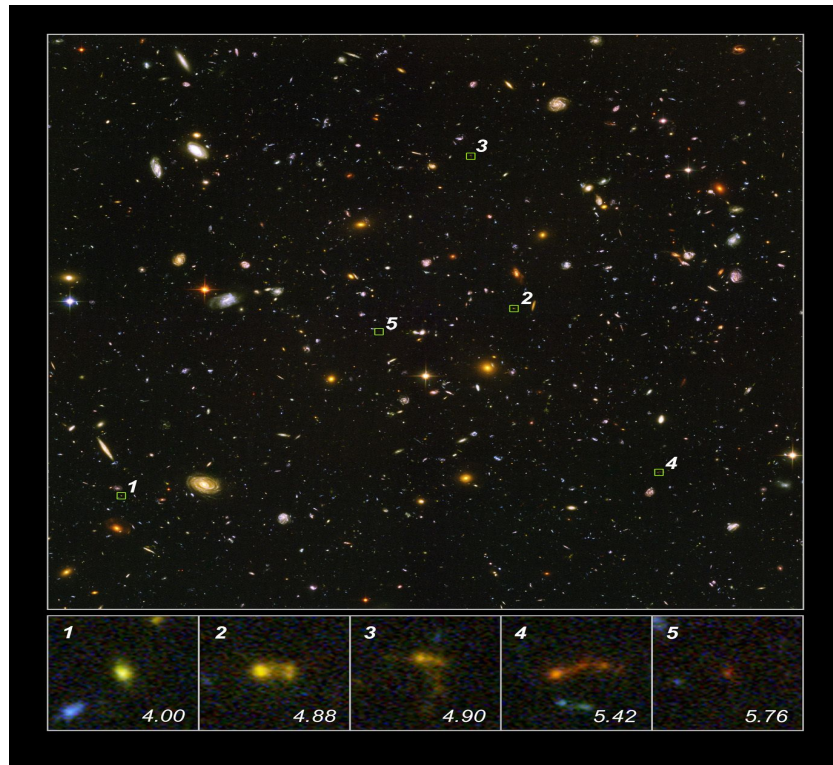
- Another key example is assessing the **sensitivity** of your method/observations
- How do you know if your observations or analysis techniques are **sensitive** enough to detect a population of galaxies/stars down to some threshold?



Applications for random sampling

Sensitivity characterization

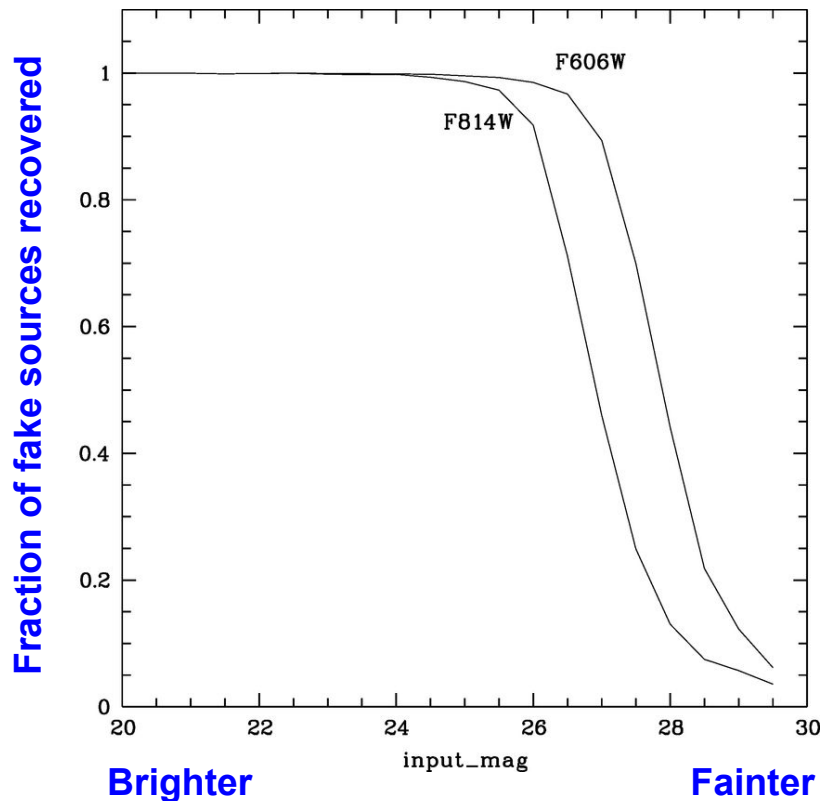
- Another key example is assessing the **sensitivity** of your method/observations
- **Randomly inject fake signals into your data**, and process them the same way you do with your real observations in order to see down to what **threshold** you can still detect sources



Applications for random sampling

Sensitivity characterization

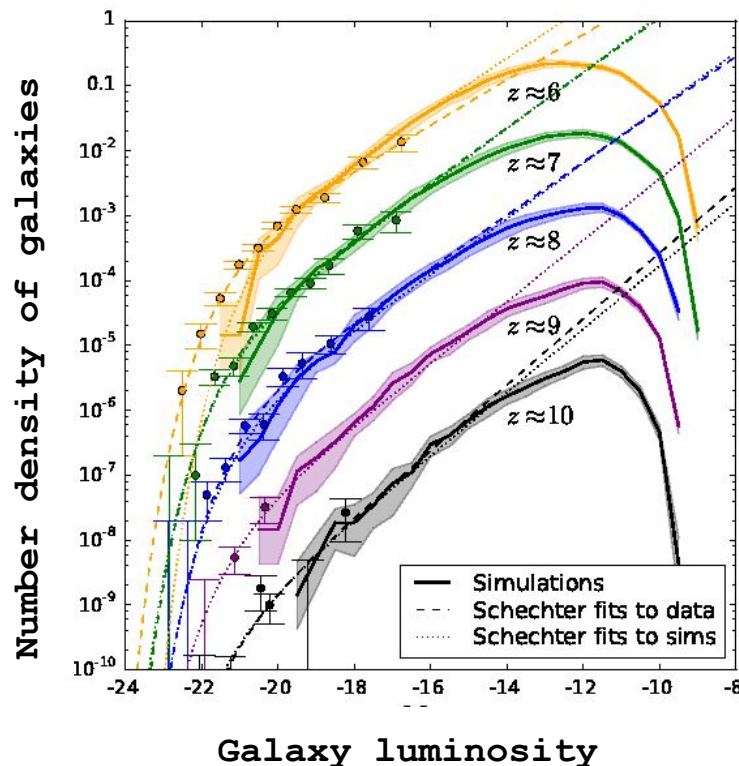
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Applications for random sampling

Sensitivity characterization

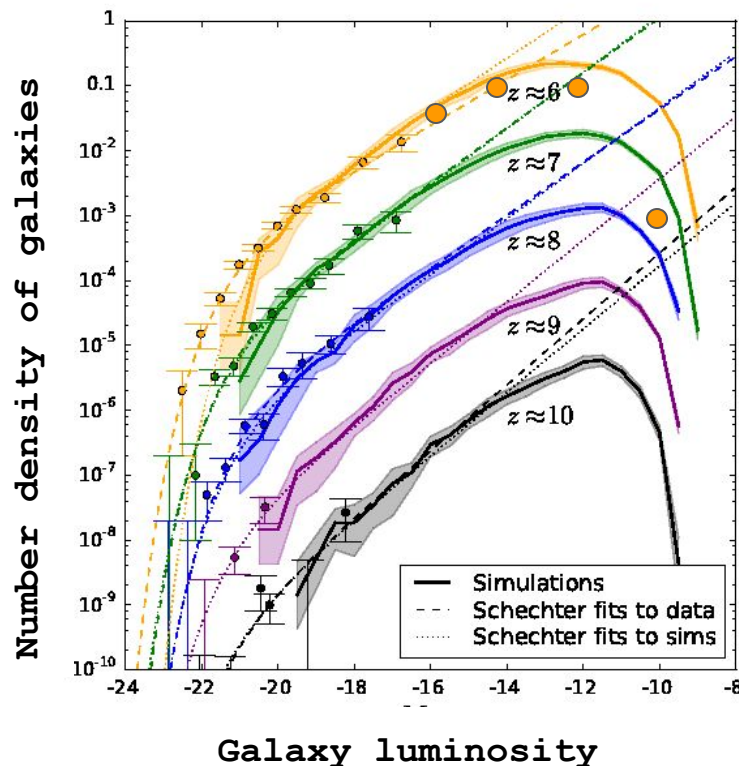
- Another example is assessing the **sensitivity** of your method/observations
- Randomly inject fake signals into your data, and see down to what threshold you can still detect them
- Extremely important when **comparing simulations with observations**



Applications for random sampling

Sensitivity characterization

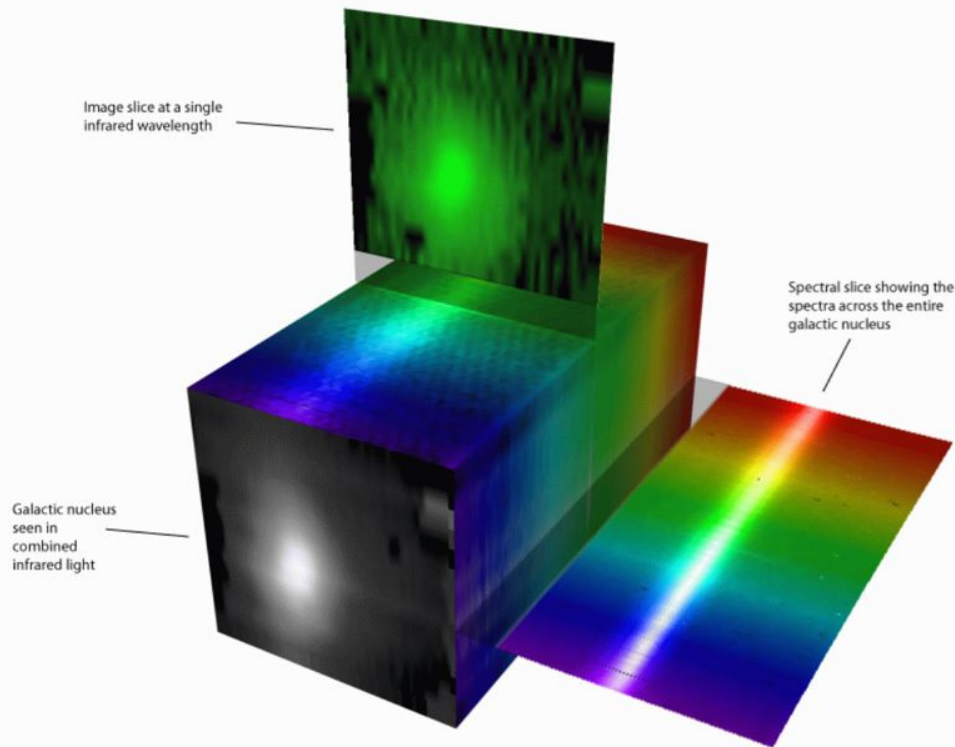
- Another example is assessing the **sensitivity** of your method/observations
- Randomly inject fake signals into your data, and see down to what threshold you can still detect them
- Extremely important when **comparing simulations with observations**



Applications for random sampling

Uncertainty propagation

- Stochastic methods for simulating data are often the *only* way to understand **error propagation** in your data analysis pipeline
- *Why?* **Non-Gaussian** and **correlated errors** are often present in data

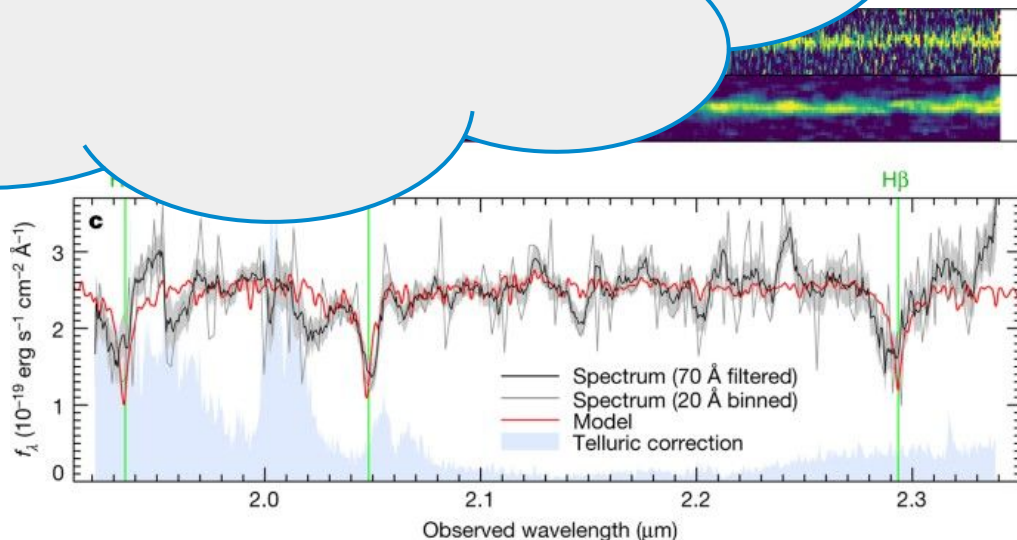


Applications for random

Uncertainty propagation

$$s_f = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 s_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 s_y^2 + \left(\frac{\partial f}{\partial z}\right)^2 s_z^2 + \dots}$$

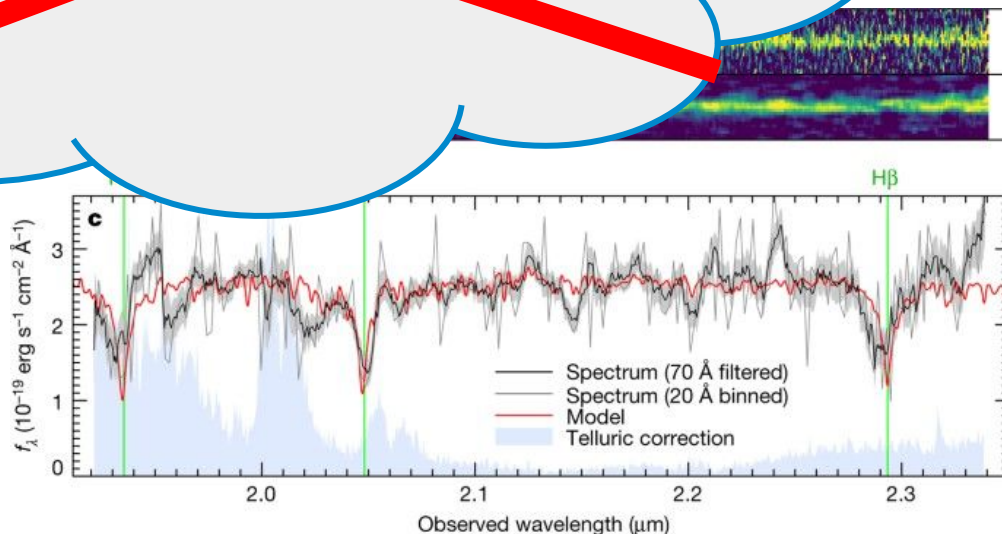
correlated errors (*no longer independent variables*) require inclusion of the full covariance matrix



Applications for random Uncertainty propagation

$$s_f = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 s_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 s_y^2 + \left(\frac{\partial f}{\partial z}\right)^2 s_z^2 + \dots}$$

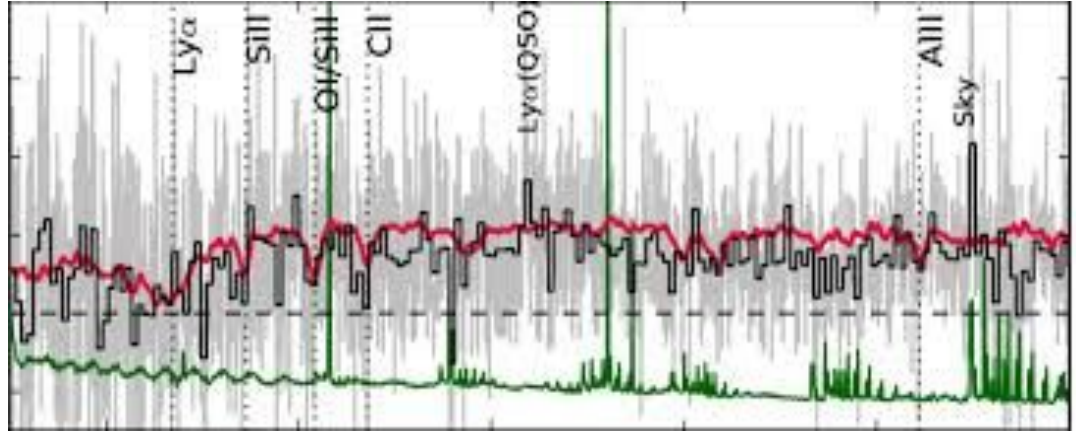
Analytic error propagation is
no longer valid, nor feasible
for an end-to-end
assessment of uncertainty



Applications for random sampling

Uncertainty propagation

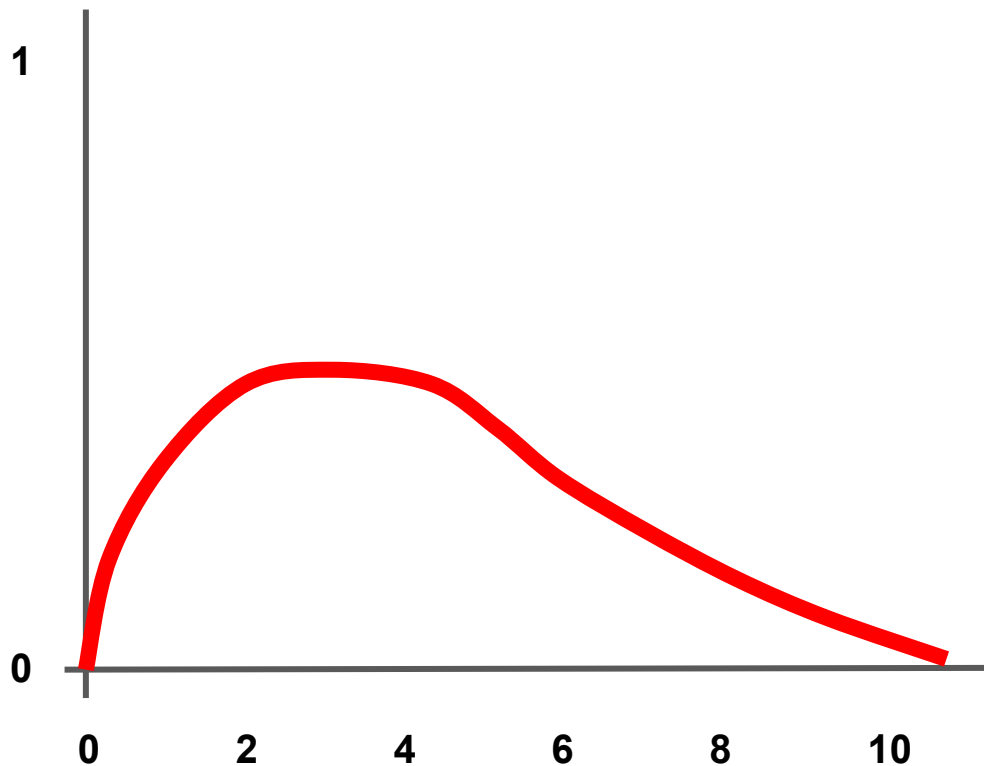
- What to do instead?
- A **numerical approach** is to generate artificial noise, or use information about uncertainties at previous steps of your analysis to **perturb** your data/model at some point in the analysis/reduction procedure, and **repeat** subsequent analysis steps many times



Methods for sampling

Inverse transform sampling

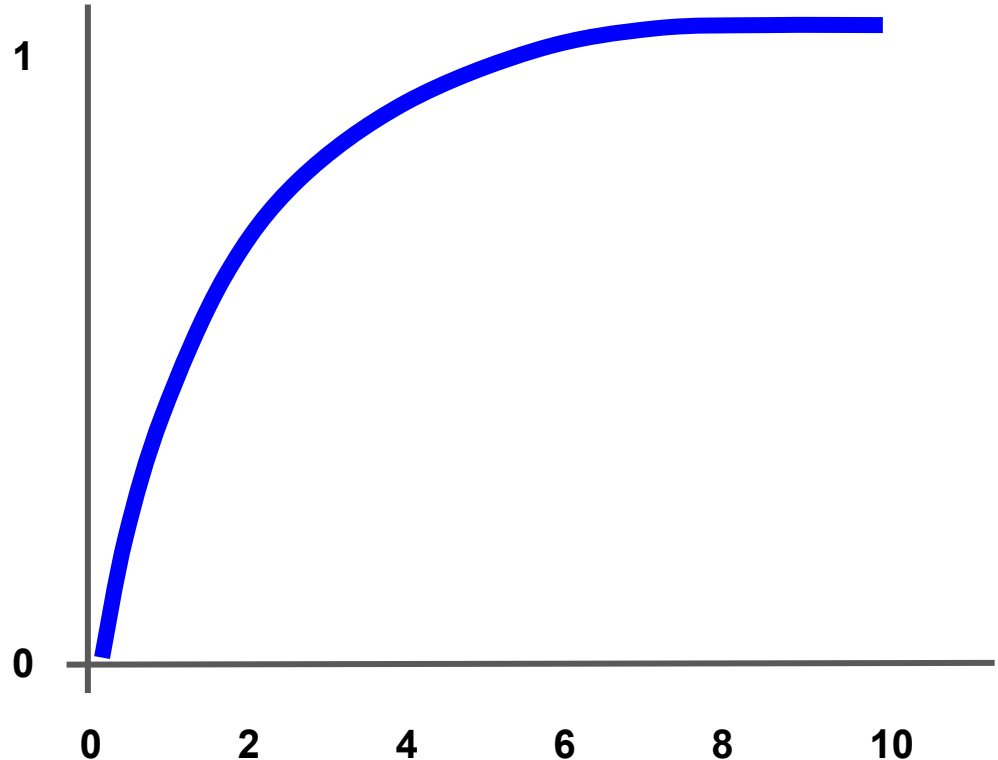
- Consider some random data which follow a probability distribution $p(x)$



Methods for sampling

Inverse transform sampling

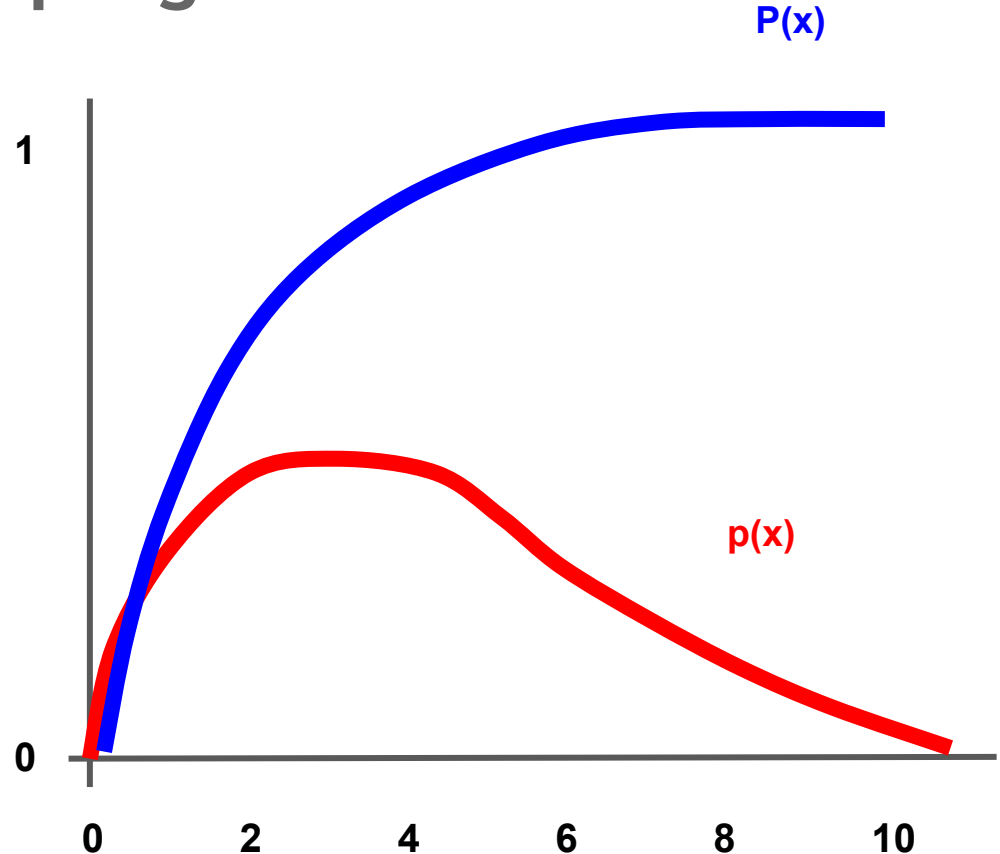
- Consider some random data which follow a probability distribution $p(x)$
- This has a corresponding cumulative distribution $P(x)$



Methods for sampling

Inverse transform sampling

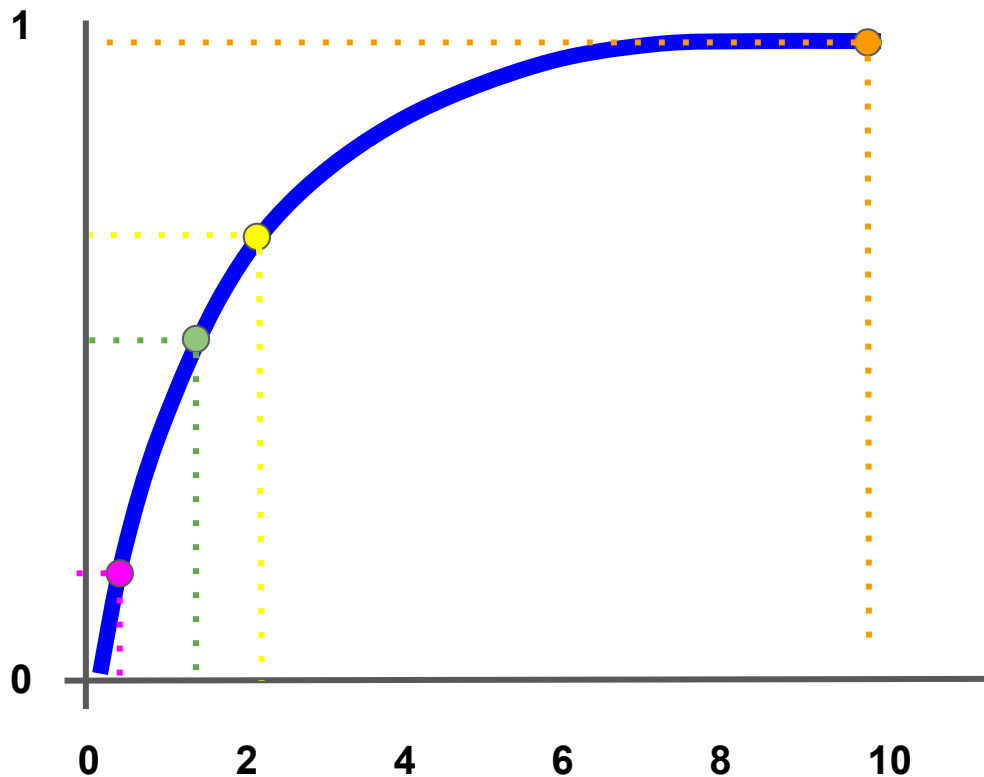
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Methods for sampling

Inverse transform sampling

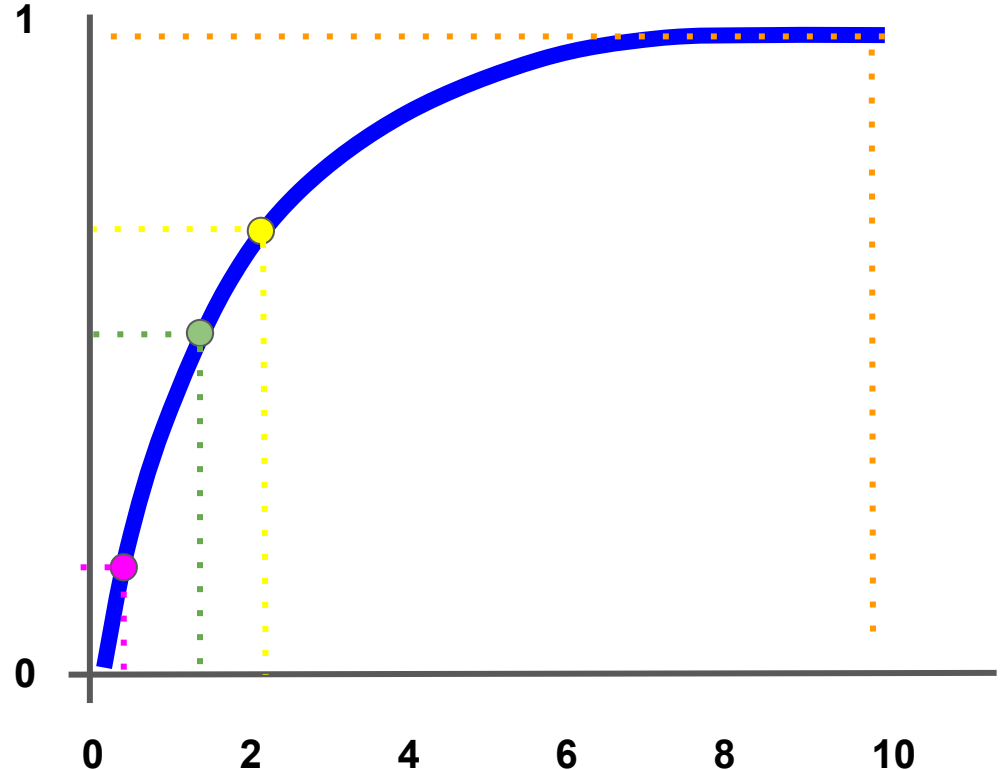
- Consider some random data which follow a probability distribution $p(x)$
- This has a corresponding cumulative distribution $P(x)$
- If we consider some uniform random number $U(0,1)$, we can ask what value x in the CDF this corresponds to.



Methods for sampling

Inverse transform sampling

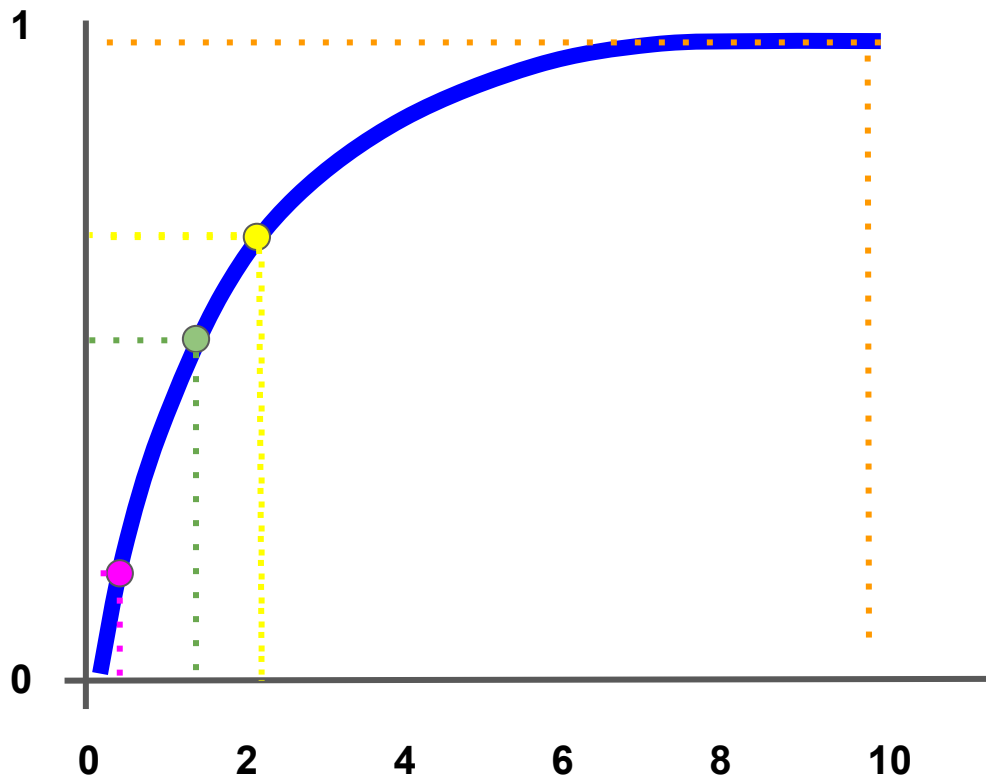
- Depending on the shape of the CDF, a uniform set of numbers from 0 to 1 will select more or fewer deviates (x-locations) at locations proportional to the underlying PDF.
- This is the ‘magic’ of inverse transform sampling which makes it very useful in many applications.



Methods for sampling

Inverse transform sampling

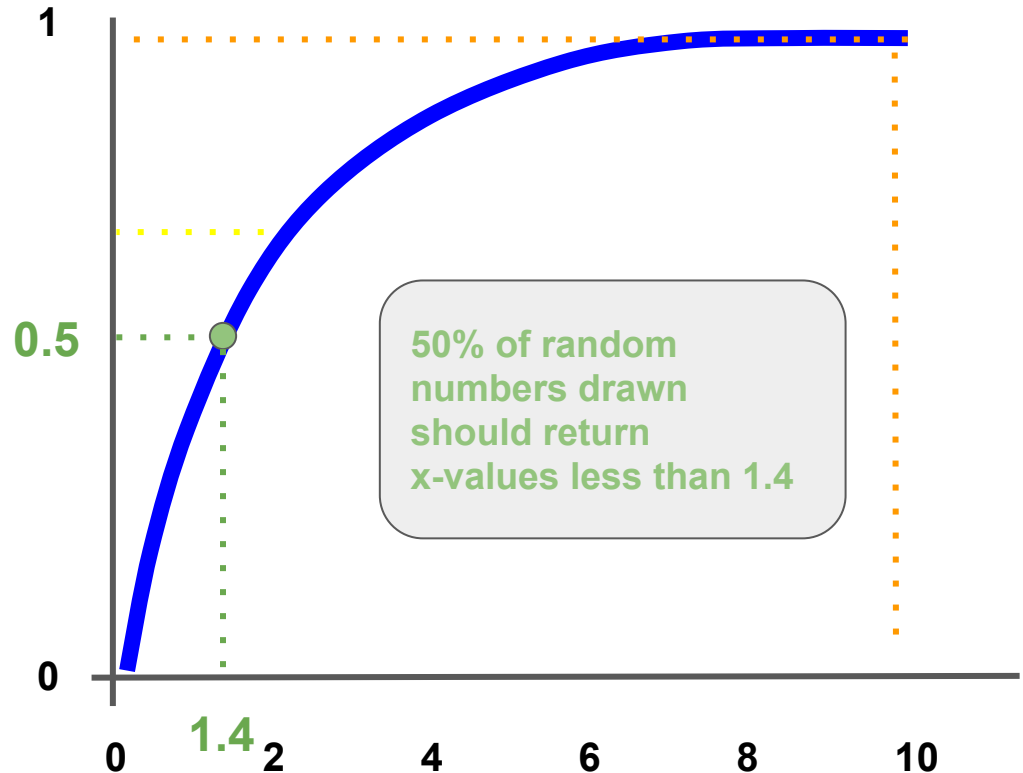
- Depending on the shape of the CDF a uniform draw of numbers from 0 to 1 will select more or fewer deviates (x-locations) at locations proportional to the underlying PDF.
- In practice you draw random numbers between 0 to 1 and then ask what x-value they intersect on the CDF curve



Methods for sampling

Inverse transform sampling

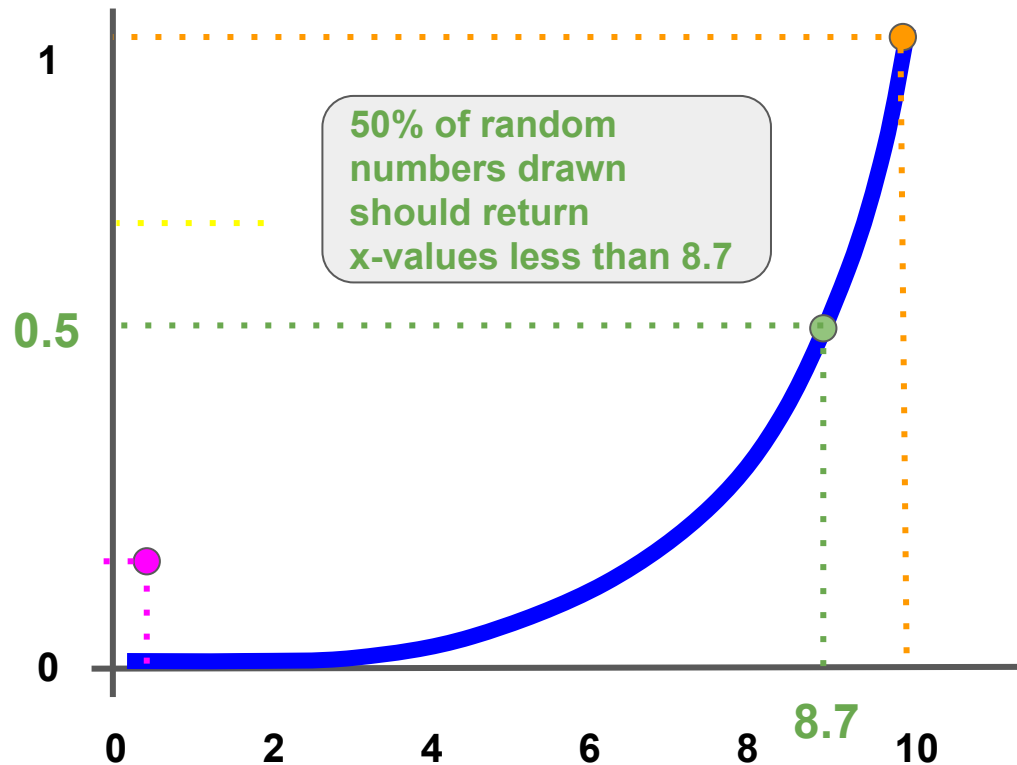
As those random numbers are uniformly distributed, the shape of the CDF determines how the resulting x-values are distributed.



Methods for sampling

Inverse transform sampling

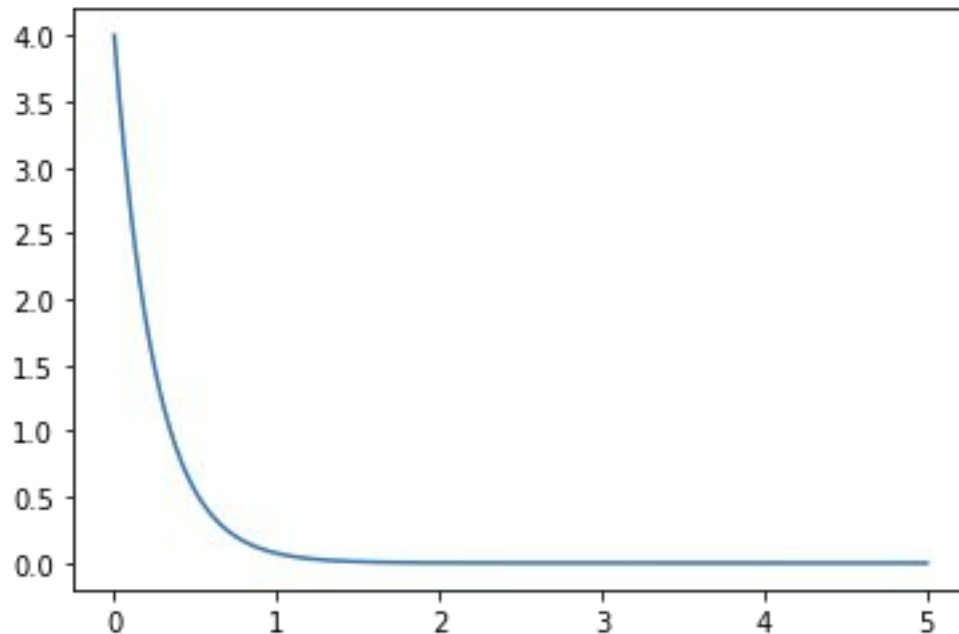
- As those random numbers are uniformly distributed, the shape of the CDF determines how the resulting x-values are distributed.
- **This gives a great way to generate random numbers from arbitrary distributions!**



Methods for sampling

Inverse transform sampling

- For a continuous distribution *sometimes* this inversion can be done analytically.
- Consider exponential distribution with PDF:
 $p(x) = 4e^{-4x}$

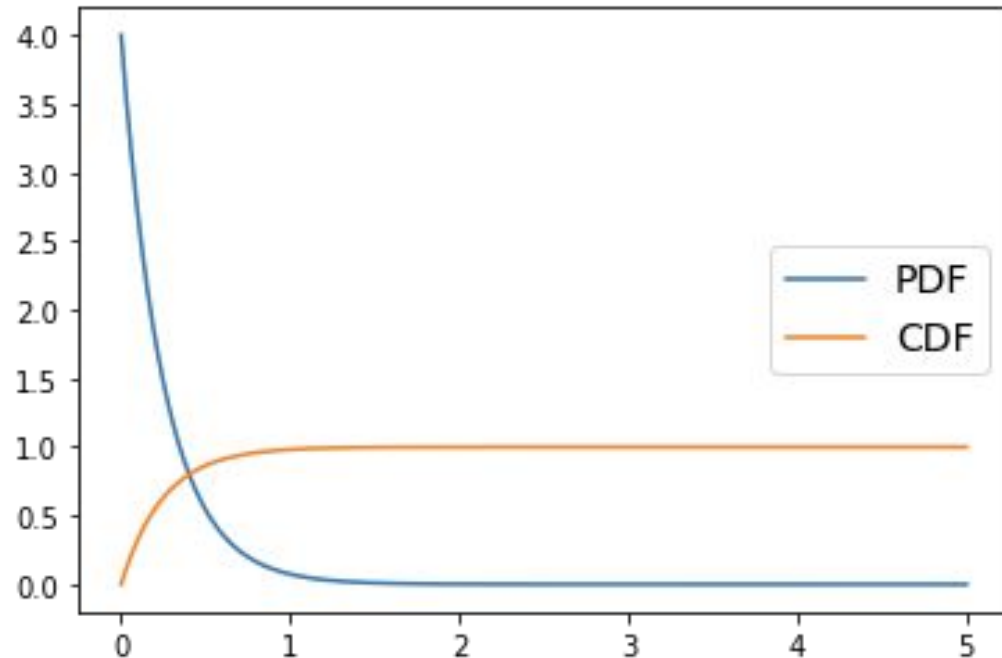


Methods for sampling

Inverse transform sampling

- For a continuous distribution sometimes this inversion can be done analytically.
- This has CDF of

$$P(y < x) = \int_0^x 4e^{-4y} dy = 1 - e^{-4x}$$

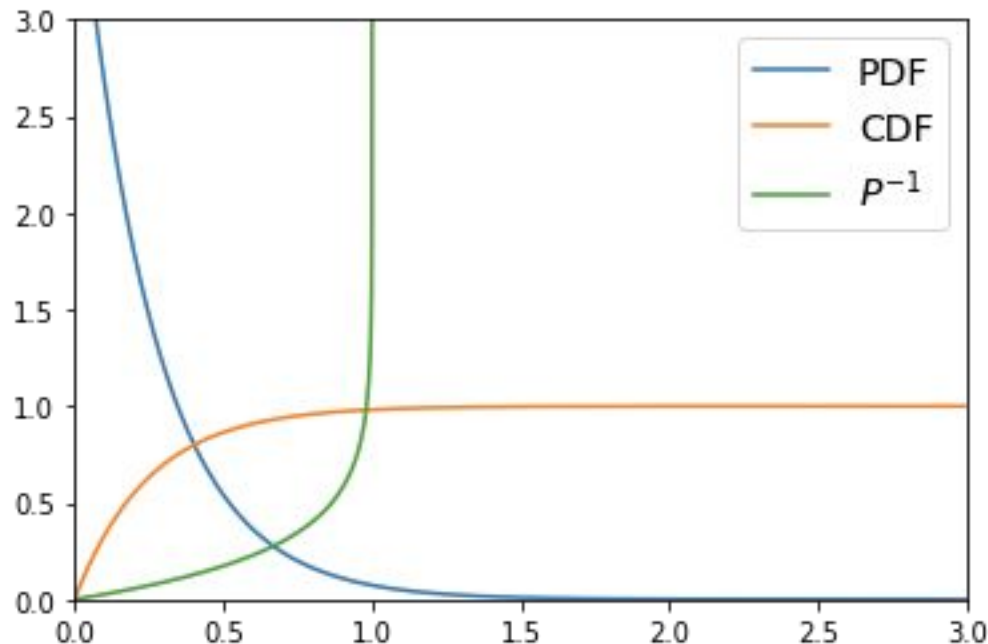


Methods for sampling

Inverse transform sampling

- For a continuous distribution sometimes this inversion can be done analytically.
- Can solve for the inverse of the CDF, $P^{-1}(y)$

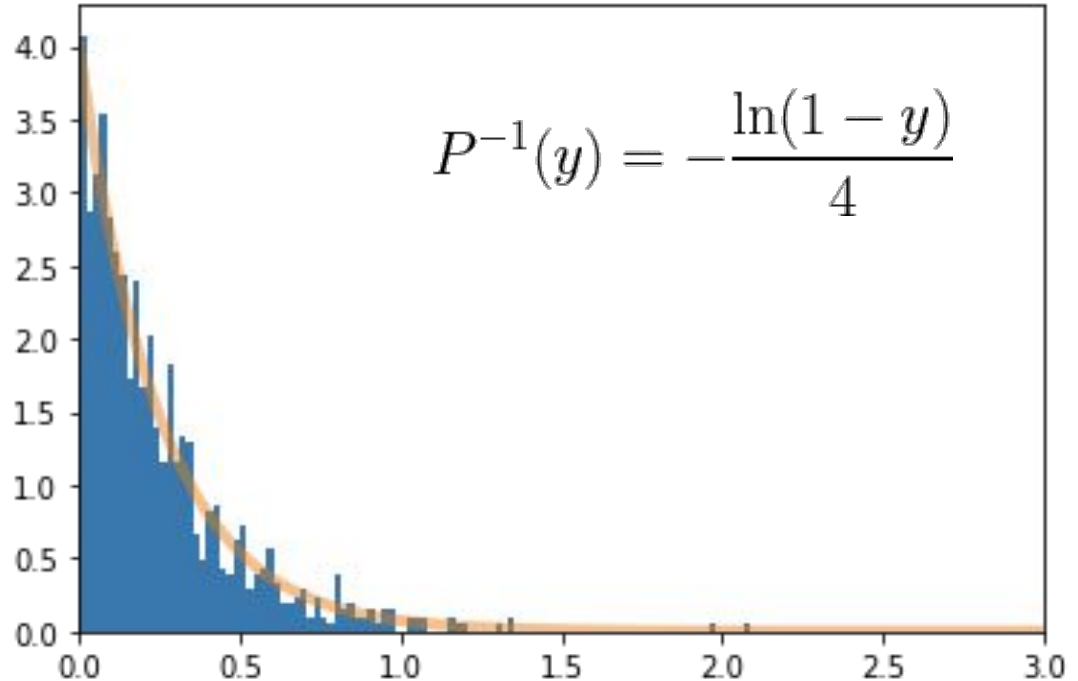
$$P^{-1}(y) = -\frac{\ln(1 - y)}{4}$$



Methods for sampling

Inverse transform sampling

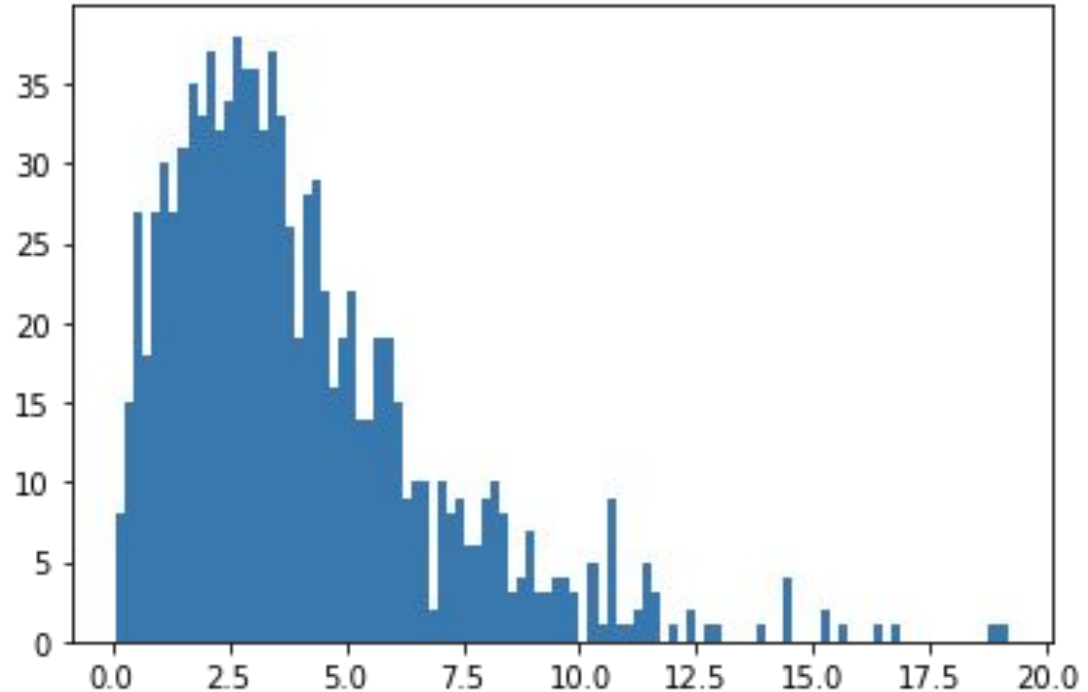
- For a continuous distribution sometimes this inversion can be done analytically.
- In these analytic cases **uniformly sampling $y = U(0,1)$ and inserting into $P^{-1}(y)$ generates your desired random distribution**



Methods for sampling

Inverse transform sampling

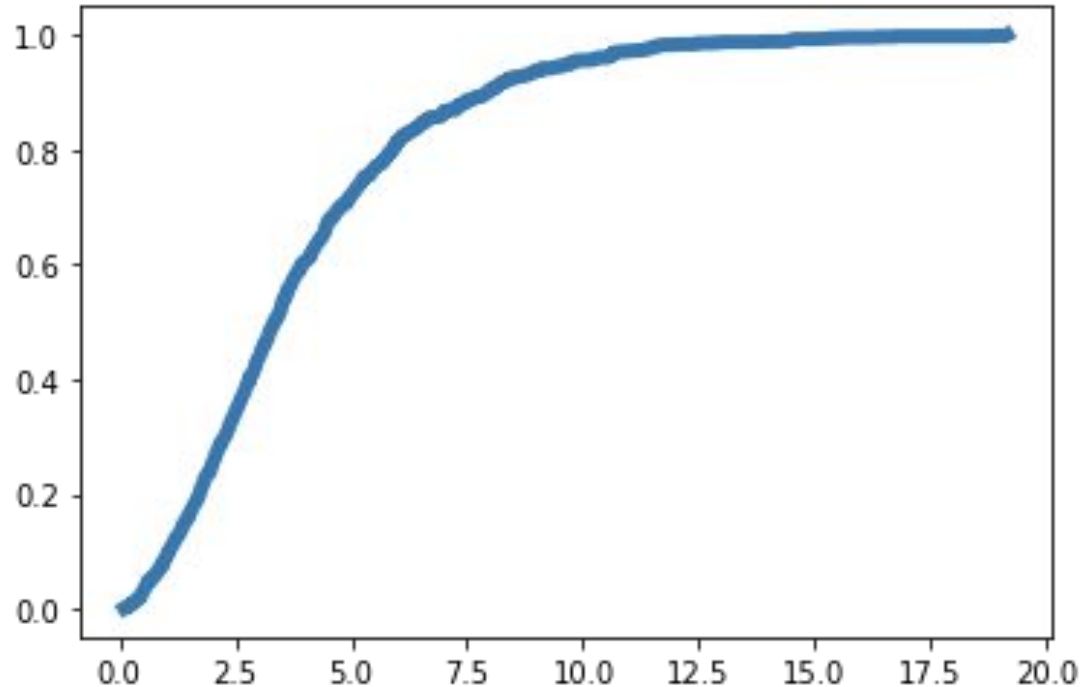
- If the CDF is not analytically invertible, or if you have discrete data, one way to proceed is as follows
- Consider the discrete data here which are roughly following a gamma distribution



Methods for sampling

Inverse transform sampling

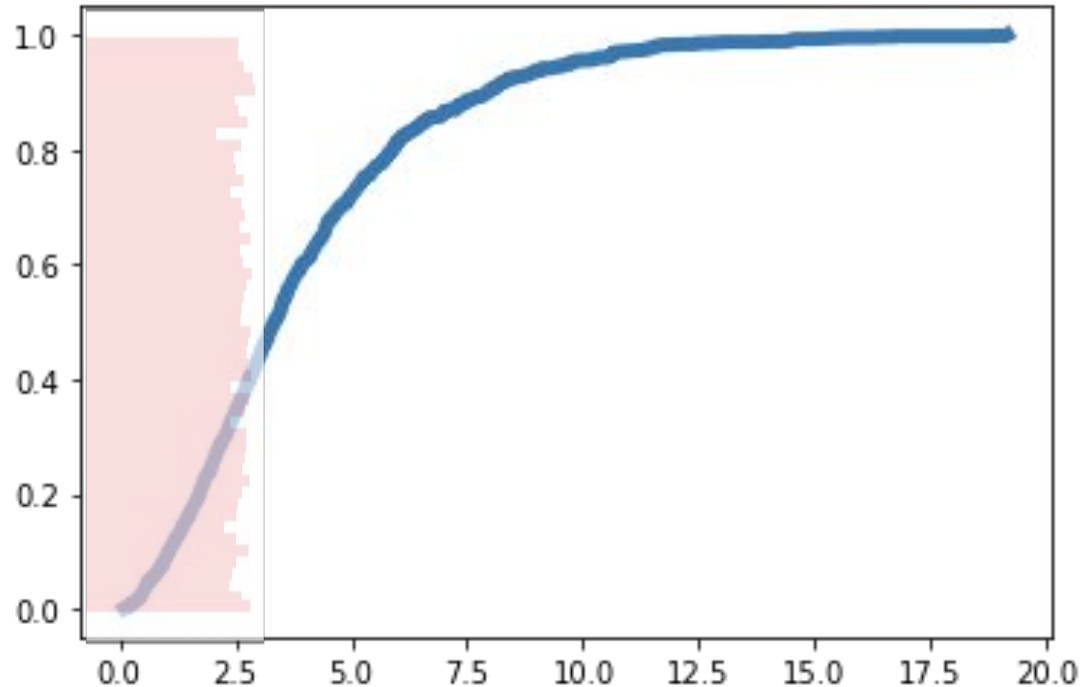
- If the CDF is not analytically invertible, or if you have discrete data, proceed as follows
- 1) Compute the discrete/empirical cumulative distribution function (e.g., a normalized cumulative sum as an approximation to the integral of an unknown PDF)



Methods for sampling

Inverse transform sampling

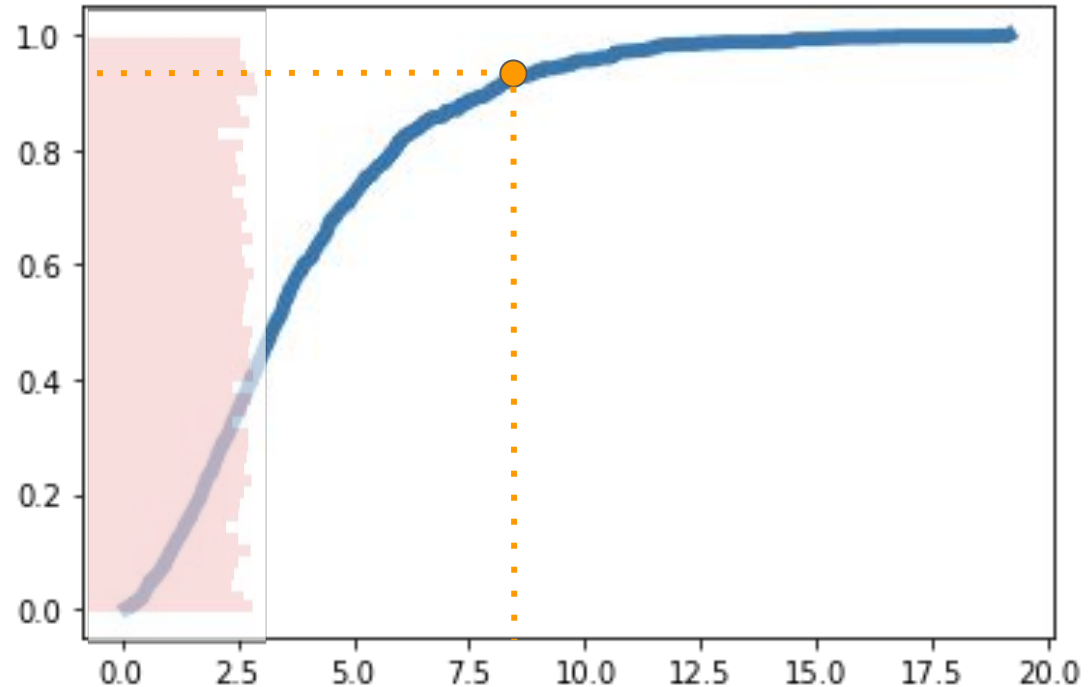
- If the CDF is not analytically invertible, or if you have discrete data, proceed as follows
- 2) Generate random uniform deviates $y=U(0,1)$



Methods for sampling

Inverse transform sampling

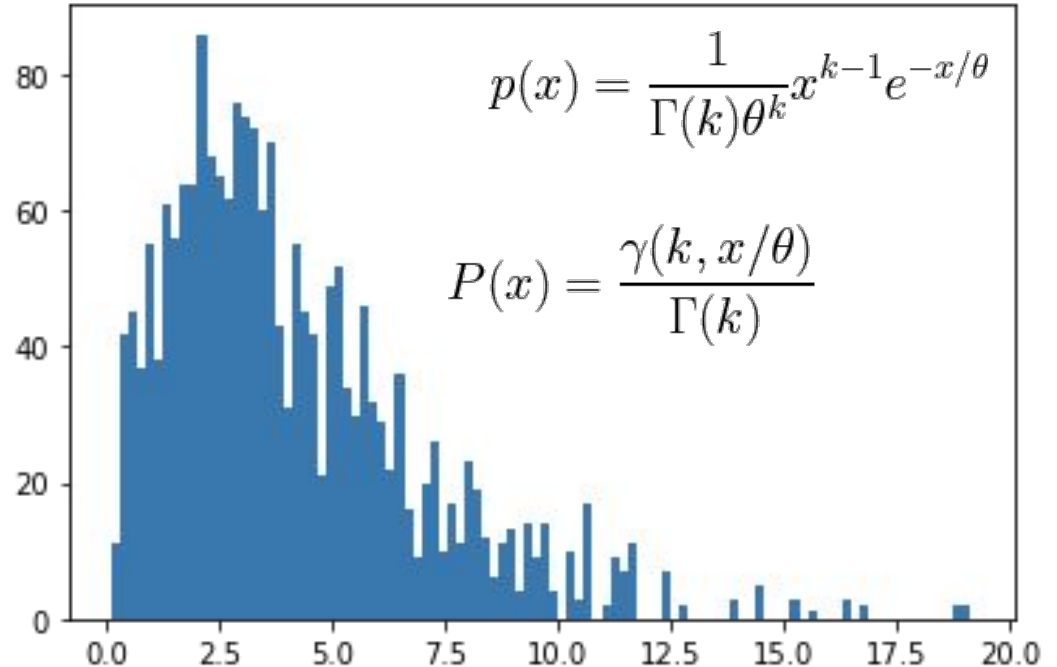
- If the CDF is not analytically invertible, or if you have discrete data, proceed as follows
- 3) Find closest location on CDF ($P(x)$) corresponding to y_i



Methods for sampling

Inverse transform sampling

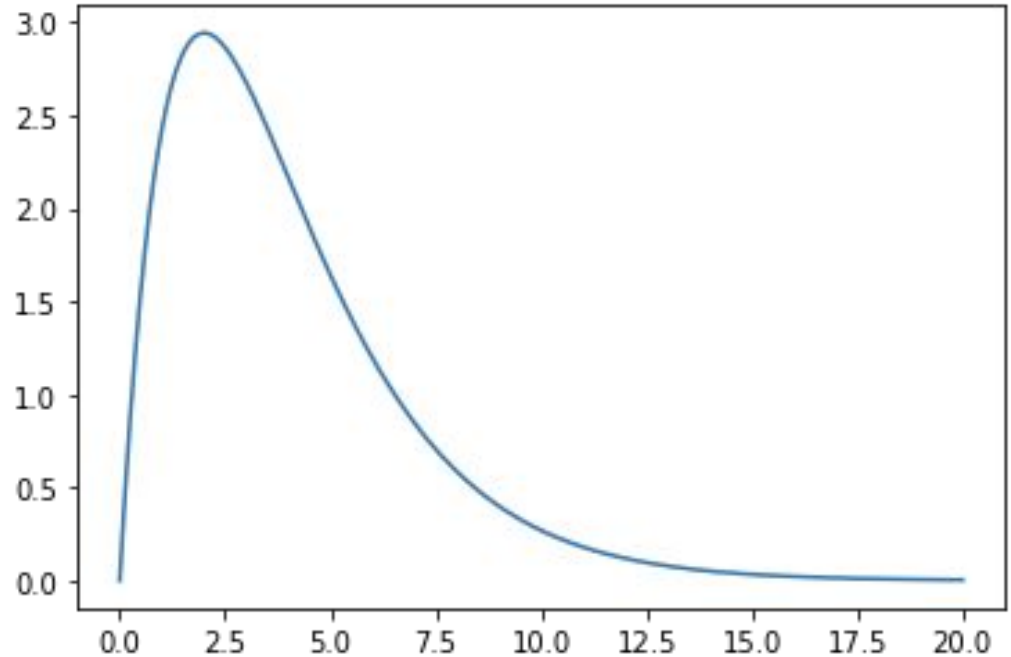
- If the CDF is not analytically invertible, or if you have discrete data, proceed as follows
- 4) Record corresponding x_i



Methods for sampling

Acceptance-Rejection sampling

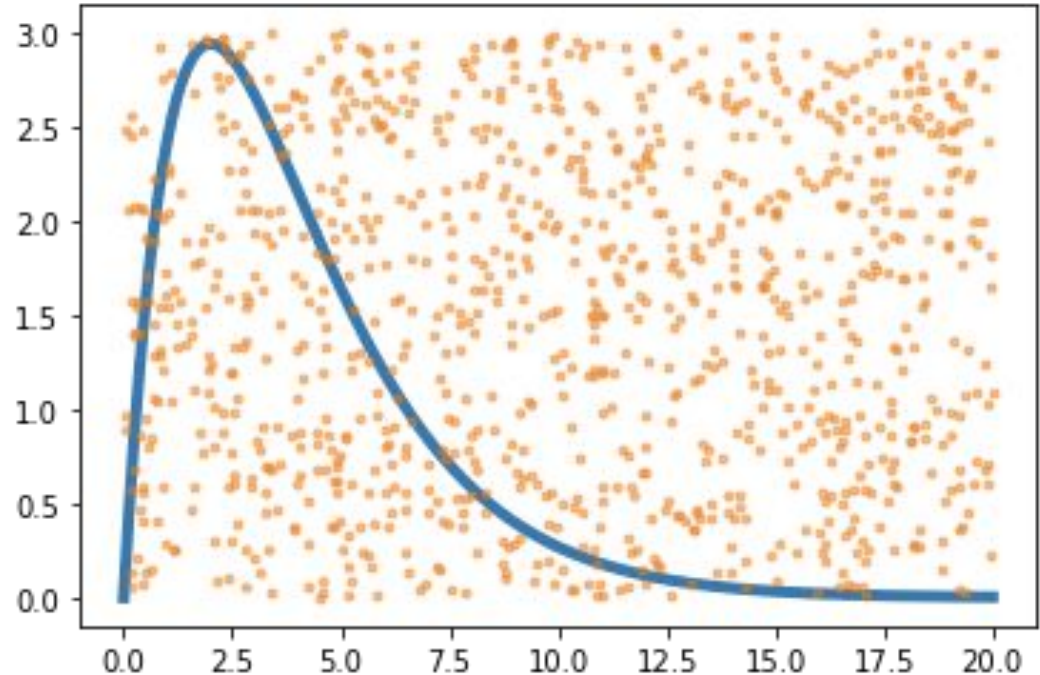
- Another method called acceptance-rejection sampling can also be used for arbitrary continuous or discrete distributions



Methods for sampling

Acceptance-Rejection sampling

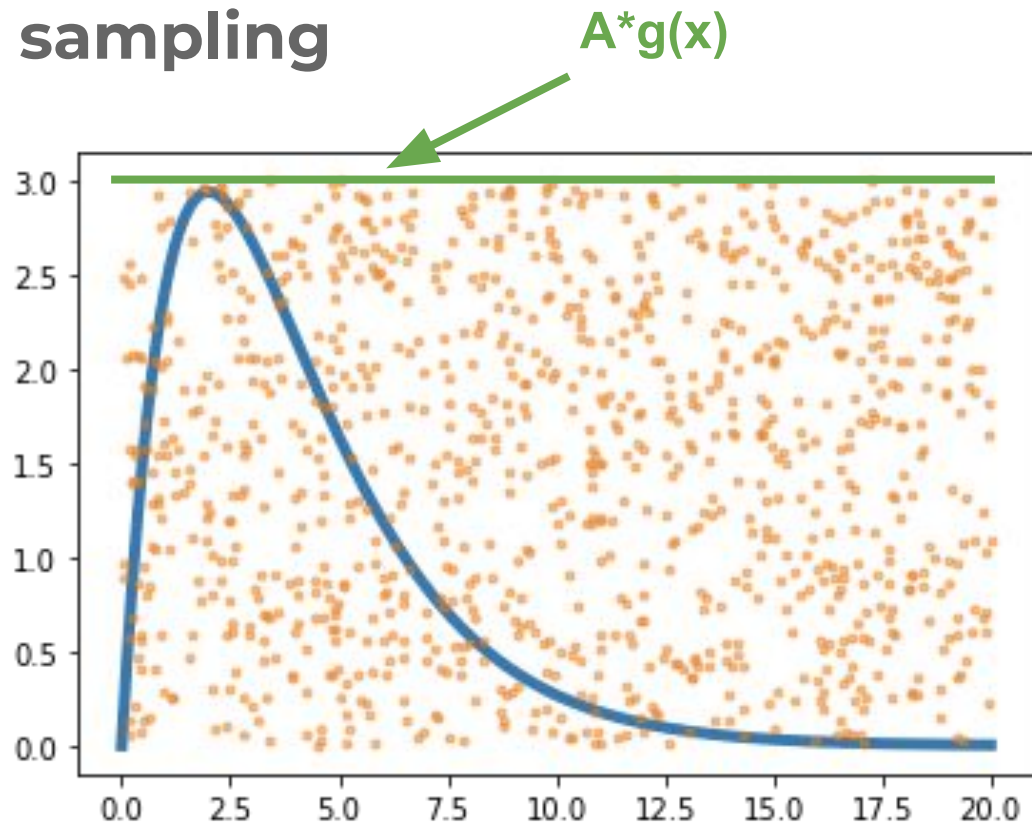
- Similar to MC integration, to evaluate the density distribution $p(\mathbf{x})$ we want to randomly sample x , and y for some values of a function $g(\mathbf{x})$ which is normalized by A to be larger than the peak of $p(x)$



Methods for sampling

Acceptance-Rejection sampling

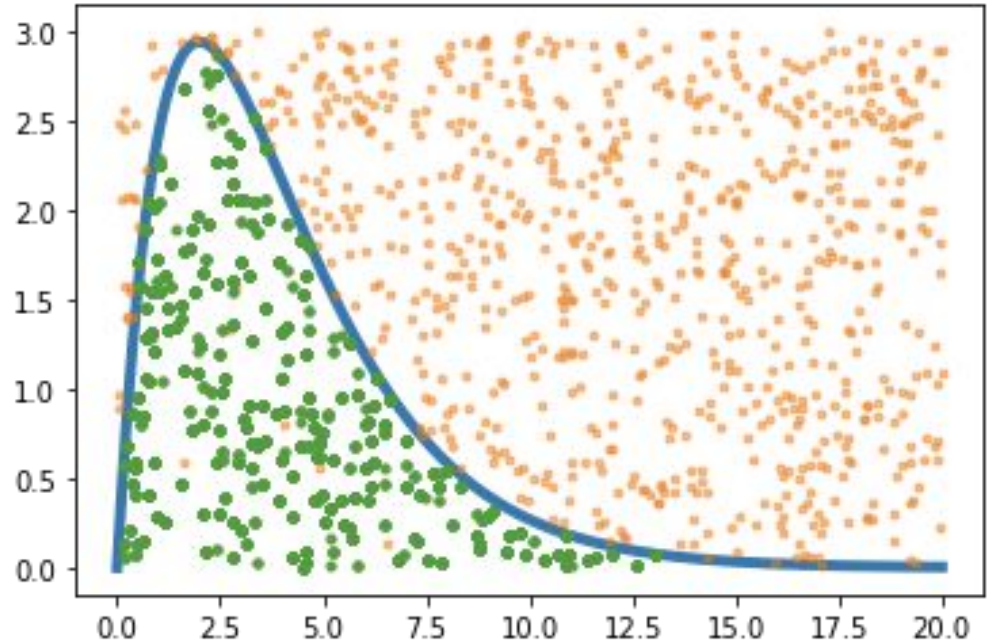
- In the simplest case consider $g(x)$ to be uniform over $[x_{\min}, x_{\max}]$ with value $A > \max(p(x))$



Methods for sampling

Acceptance-Rejection sampling

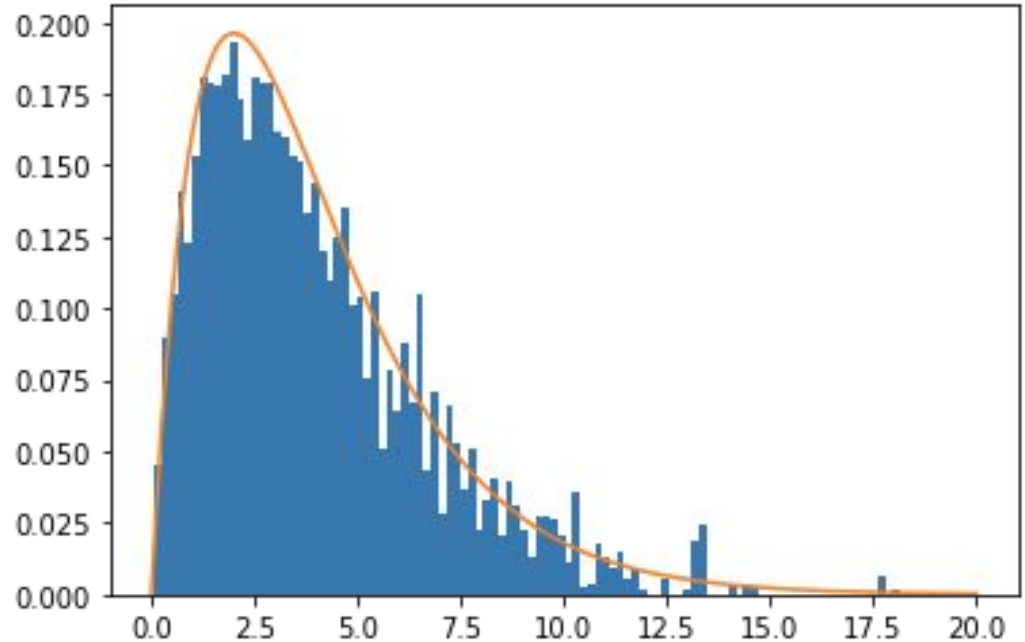
- For each of your random deviates (x_r, y_r) , retain the x_r samples where the corresponding y_r is less than $p(x_r)$



Methods for sampling

Acceptance-Rejection sampling

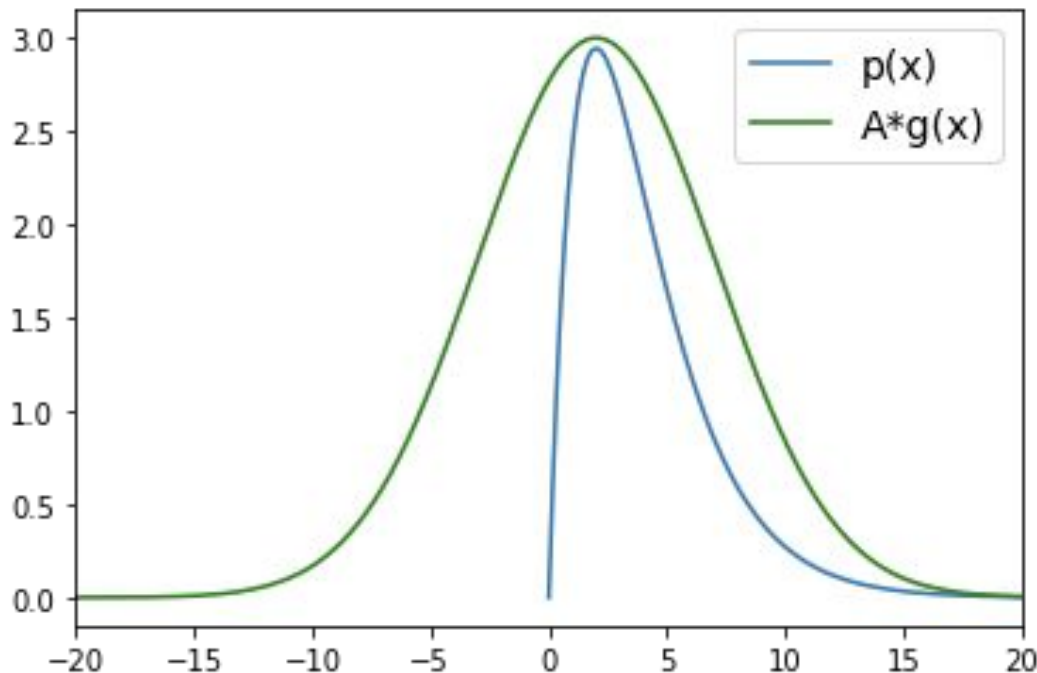
- The distribution of these successful x_r should reproduce the distribution
- If you desire an *exact number* of random deviates, this method can be less useful than inverse transform sampling.



Methods for sampling

Acceptance-Rejection sampling

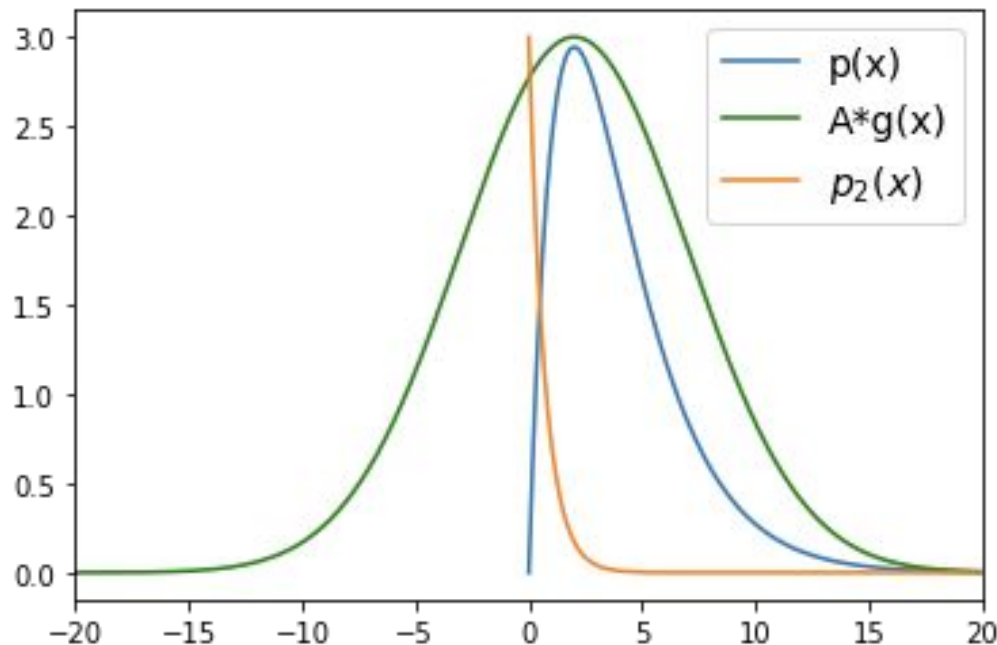
- If your target density distribution $p(x)$ is very sparse you may want to consider something other than uniform for $g(x)$



Methods for sampling

Acceptance-Rejection sampling

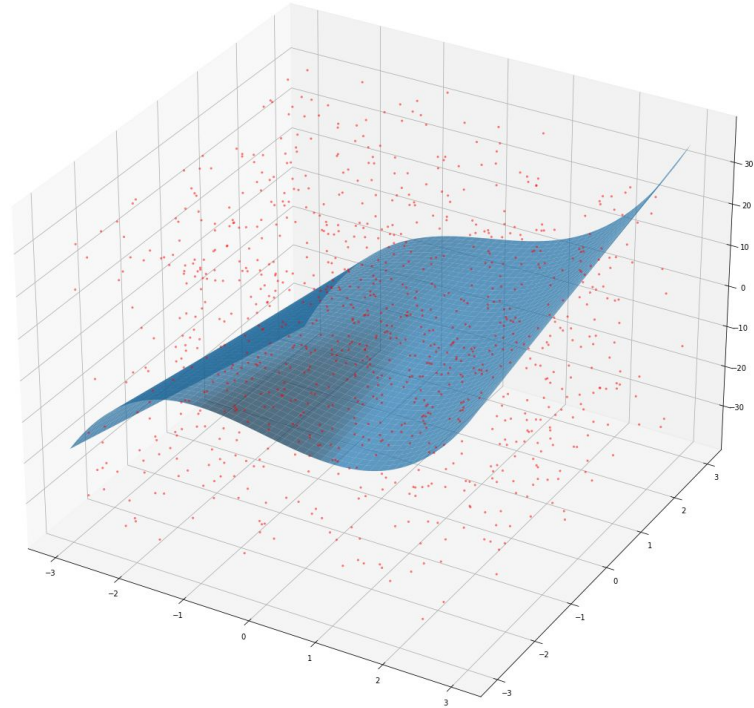
- If your target density distribution $p(x)$ is very sparse you may want to consider something other than uniform for $g(x)$



Methods for sampling

Acceptance-Rejection sampling

- Can be extended to higher dimensional or conditional distributions.
- For example, sample and accept according to the PDF of x and then reject additional points which don't satisfy the PDF of $y, z, \dots$



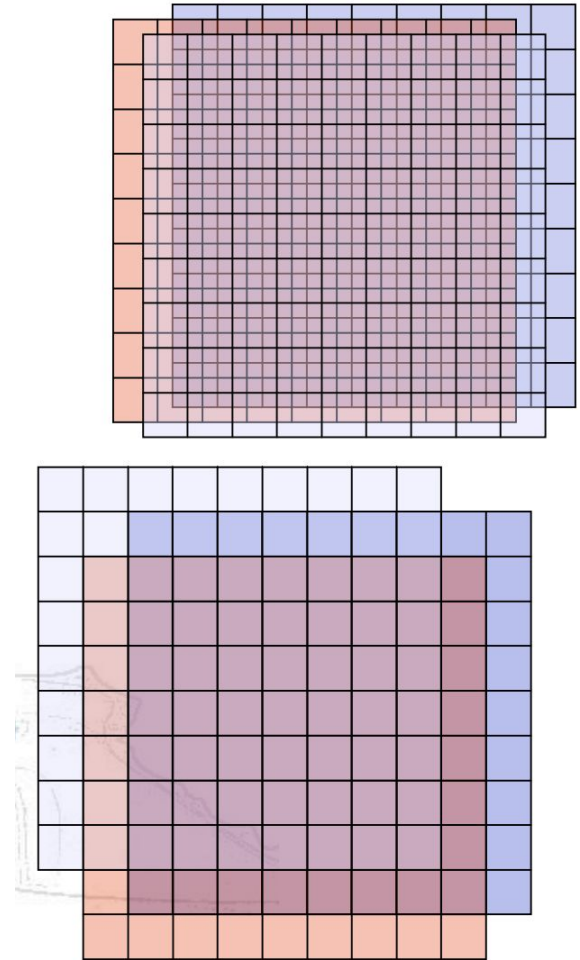
Methods for sampling

Summary

- Both **inverse transform sampling** and **acceptance-rejection sampling** offer a way to produce random numbers following some arbitrary distribution.
- You can use these techniques to calculate the following:
 - * Discrete data which you want to re-sample, reproduce or simulate
 - * An analytic form to the CDF with which to generate mock data
 - * An analytic form to the PDF you want to generate mock data from

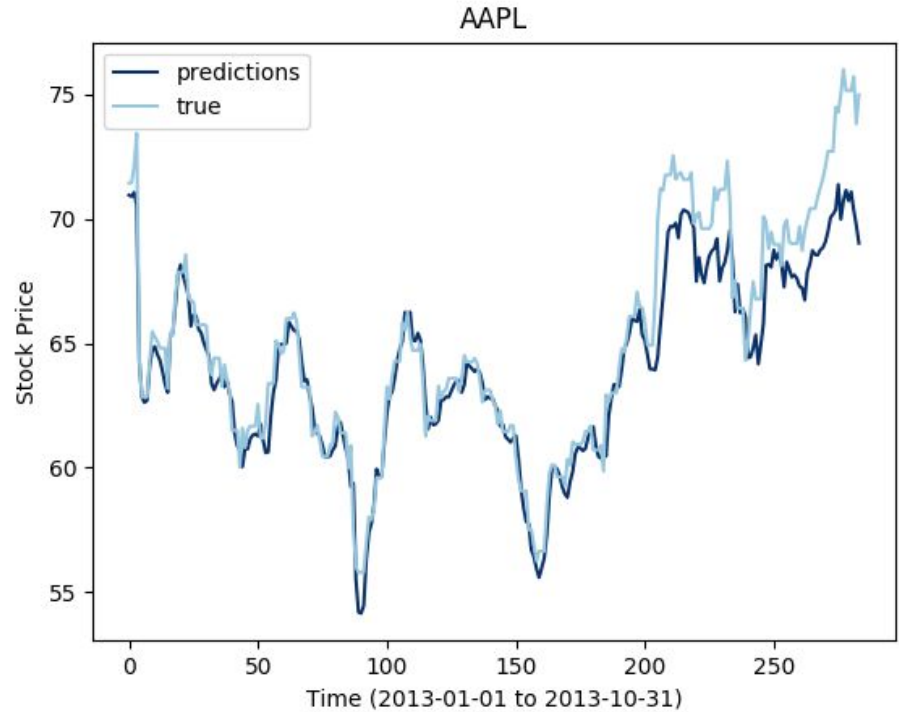
Noise characteristics

- Often want to inject or characterize noise in your data, or mock data
- The level of correlation of this noise can be important
- We often assume noise will be Gaussian and uncorrelated but this can break down frequently



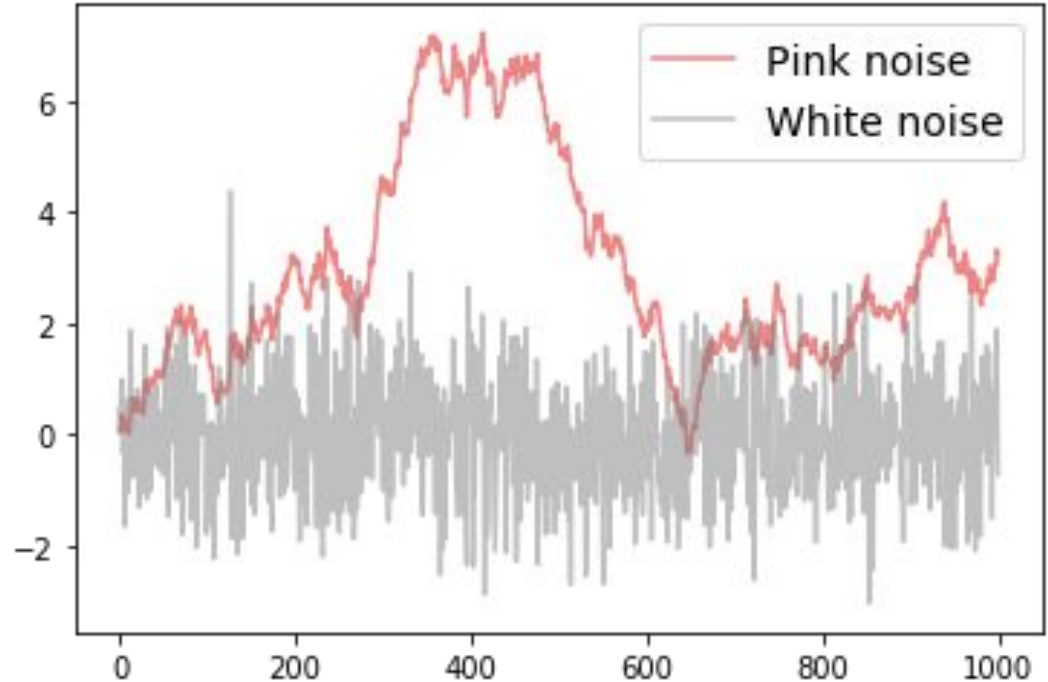
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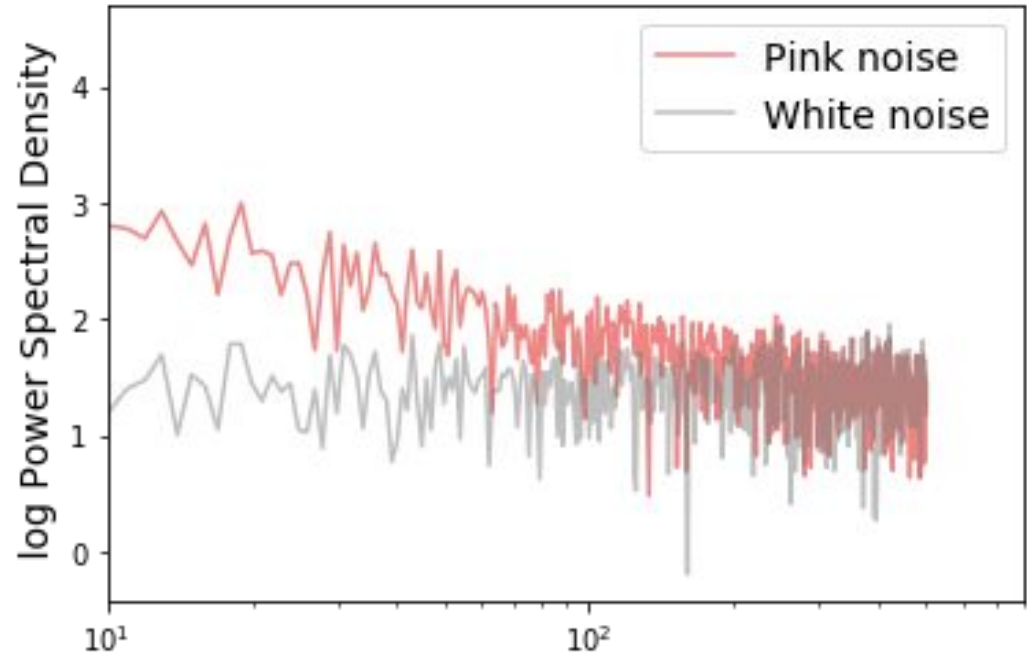
Noise characteristics

- For practical purposes can consider white noise as noise that is perfectly uncorrelated (often is, but does not necessarily have to be Gaussian)
- Pink noise is an example of noise that is correlated



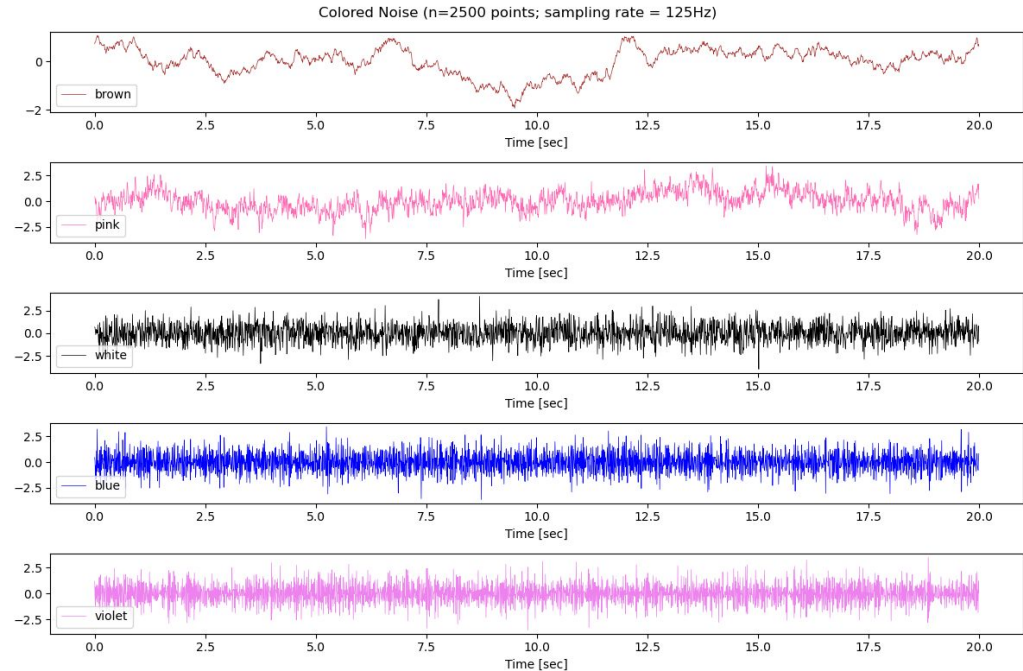
Noise characteristics

- *If* this characteristic is suitable for your problem, pink noise can be approximately generated by taking the cumulative distribution of the white noise series and normalizing by the sum.



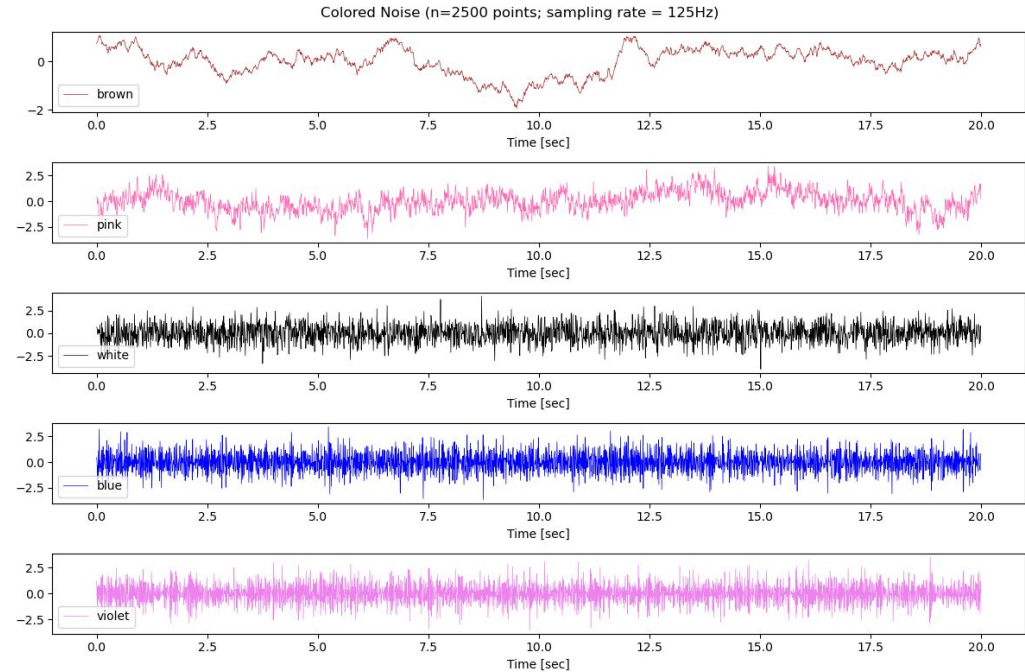
Noise characteristics

- Other noise power spectra can be produced using the `colorednoise` package in python.
- These can generate noise with power-law indices for a given spectral density of arbitrary values.
- This might be helpful when trying to forward model instrumental effects where you have correlated noise (e.g., spectra, IFU cubes)



Noise characteristics

- If you have desired characteristics/correlations for your noise that are different than white/brownian/pink we have included a routine to add noise with different levels of correlation for you to experiment with.



Python random number generation recap

- The preceding and later slides and algorithms help provide ways to do arbitrary sampling.
- However throughout these keep in mind what their limitations are with respect to any of the optimization or other applications we discussed,
- As a reminder *numpy.random* can provide random samples from a few useful PDFs.

`beta(a, b[, size])` `binomial(n, p[, size])` `chisquare(df[, size])` `dirichlet(alpha[, size])`
`exponential([scale, size])` `f(dfnum, dfden[, size])` `gamma(shape[, scale, size])`
`geometric(p[, size])` `gumbel([loc, scale, size])` `hypergeometric(ngood, nbad, nsample[, size])`
`laplace([loc, scale, size])` `logistic([loc, scale, size])` `lognormal([mean, sigma, size])`
`logseries(p[, size])` `multinomial(n, pvals[, size])` `multivariate_normal(mean, cov[, size, ...])`
`negative_binomial(n, p[, size])` `noncentral_chisquare(df, nonc[, size])`
`noncentral_f(dfnum, dfden, nonc[, size])`
`normal([loc, scale, size])` `pareto(a[, size])` `poisson([lam, size])` `power(a[, size])`
`rayleigh([scale, size])` `standard_cauchy([size])` `standard_exponential([size])`
`standard_gamma(shape[, size])`
`standard_normal([size])` `standard_t(df[, size])` `triangular(left, mode, right[, size])`
`uniform([low, high, size])` `vonmises(mu, kappa[, size])` `wald(mean, scale[, size])`
`weibull(a[, size])` `zipf(a[, size])`